



A novel hybridized feature selection strategy for the effective prediction of glaucoma in retinal fundus images

Law Kumar Singh¹ · Munish Khanna² · Shankar Thawkar³ · Rekha Singh⁴

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Abstract

Feature selection (FS) is crucial to transforming high-dimensional data into low-dimensional data. The FS approach selects influential traits and ignores the rest. This approach improves machine learning (ML) classifiers by reducing computational complexity and solution time. This empirical study presents a novel and effective methodology that uses two contemporary state-of-the-art soft-computing algorithms, the Grey Wolf Optimizer (GWO) and the Whale Optimization Algorithm (WOA). We have also created the hybrid version (hGWWO) of these two approaches as our novel, innovative scientific contribution. The baseline algorithms above have been used previously for feature selection across different domains. According to our understanding, these three algorithms are being used for the first time in glaucoma identification, particularly on the publicly available benchmark dataset, ORIGA. The rising global prevalence of glaucoma prompted this proposed methodology's focus on the illness. This illness is second only to cataracts in causing visual loss. Medical imaging professionals are examining retinal scans to diagnose glaucoma. Manual eye screening and retinal fundus imaging for confirmation of this infection require skilled ophthalmologists. The screening analysis method is time-consuming, requires experienced staff, and is subject to observational differences. In order to overcome these issues and to support the medical fraternity, an artificial intelligence-supported computer-aided clinical decision support system (CA-CDSS) is implemented in the present endeavor for confirmation of this disease from retinal fundus images. Nature-inspired computing strategies for feature selection and ML models for classification are employed to classify fundus retinal images under investigation. From the ORIGA dataset, sixty-five features were retrieved. A subset of most influential features is selected from the original dataset using three soft-computing-based FS methods. ML classifiers are trained using this portion of data and evaluated using a 70:30 technique. The suggested method yielded 96.8% accuracy, 0.981 specificity, 0.992 sensitivity, 0.969 precision, and a 0.982 F1-score. This study shows fresh initiatives with positive effects on ophthalmologists, researchers, and the public.

Keywords Glaucoma prediction · Retinal fundus images · Grey wolf algorithm · Whale optimization algorithm · Hybrid approach

1 Introduction

In recent years, notable progress has been made in high-throughput technologies, specifically in the realm of cancer recurrence or disease occurrence management. The aforementioned advancements have led to the generation of diverse forms of high-dimensional data [1]. Every individual piece of medical information holds inherent value that is distinct and significant in its own right. The increased dimensionality of the data presents a greater level of intricacy in the process of drawing conclusions from it. There exists an urgent need to employ dimension reduction methodologies in order to convert data with a high number of dimensions into a representation with a reduced number of dimensions. The process of data reduction enables the effective management of multidimensional data by employing various techniques, including categorization, presentation, transmission, and storage. Based on prior research [2], inadequate data quality can manifest in various ways, including the existence of noise, anomalies, data scarcity or duplication, and data that lacks representativeness or demonstrates bias. Furthermore, the process of dimensionality reduction has the potential to eliminate features that are not utilized. However, it is important to note that this procedure may potentially lead to a decrease in the number of interconnected features. As a result, this reduction can contribute to the mitigation of noise and ultimately facilitate the advancement of more precise learning models [5]. The implementation of the FS methodology possesses the capacity to adeptly address a multitude of challenges. One of the primary benefits of employing FS in practical applications, particularly within the medical sector, lies in its capacity to effectively reduce costs associated with unwarranted testing for pathological analysis, all the while ensuring dependable detection. This, consequently, facilitates the preservation of patients' time [3]. It is widely recognized among scholars that the limitation of input features confers advantages to the advancement of classifier models, resulting in enhanced computational efficiency and predictive efficacy. This observation highlights the importance of this distinction, as it suggests that medical diagnostic tests and evaluations with less elaborate procedures generally offer enhanced cost-effectiveness [4].

Well-known feature selection methods are used to find discriminatory attributes [6, 7]. The selection of robust features is crucial for classification algorithms to generate intelligent models. ML algorithm design and implementation depend on FS, which improves classification accuracy while conserving computational resources [8, 9]. Classifier performance may suffer from a feature space with many irrelevant or duplicated characteristics. Using feature selection algorithms, a subset of the most important attributes from the initial set is created to improve classifier accuracy by removing redundant features. Researchers have developed and implemented many bio-inspired computing technologies to reduce feature space dimensions. According to numerous studies, classification systems' accuracy depends on the careful selection of relevant characteristics [6–10]. FS methods typically select the most prominent and non-duplicative features (attributes). Two soft computing algorithms, GWO and WOA, were used to evaluate the proposed methodology's accuracy and efficacy. GWO and WOA were combined to create the third FS method (hGWWO), the fusion method. The hGWWO hybrid algorithm is a novel way to identify infection-related traits. The goal of this algorithmic approach is to find the optimal "n" attributes for distinguishing glaucomatous and normal images. It also efficiently removes redundant and unnecessary functions, improving classification accuracy and processing costs. The comprehensive, high-dimensional sixty-five feature set from the ORIGA dataset supports our proposed strategy. Based on empirical evidence, this approach can yield positive results.

This study describes the successful implementation of a CA-CDSS designed to accurately distinguish glaucomatous images. The classification was achieved using six popular ML classifiers.

The decision to select glaucoma as the focal point of investigation is based on its classification as a progressive optic neuropathy that affects an estimated population of approximately 3 million individuals residing in the Western Hemisphere. Nevertheless, it is important to acknowledge that a mere fraction of the impacted populace possesses an understanding of this particular condition. The global incidence of blindness is estimated to be around 120 million cases, with the United States contributing approximately 9% to 12% of this total. There is a notable disparity in the prevalence of visual impairment between African Americans and Caucasians, whereby African Americans exhibit a 15-fold higher likelihood of experiencing this condition. The estimated prevalence of glaucoma among individuals in India is approximately 60 million. Glaucoma is widely recognized for its substantial influence on a considerable population, impacting an estimated 12 million individuals on an annual basis and leading to approximately 1.2 million cases of visual impairment resulting in blindness [11]. The aforementioned factor plays a significant role in the probability of developing glaucoma, which is a prominent determinant in the occurrence of enduring visual impairment. The facilitation of glaucoma detection can be achieved through the utilization of fundus camera images [12]. Patients who have been diagnosed with glaucoma present with enlarged optic cups that occupy the optic disc within their ocular structures. Glaucoma, a pathological condition affecting the eyes that can result in irreversible vision impairment, often presents a dearth of discernible symptoms, thereby presenting considerable difficulties in its prompt detection. However, should the detection of glaucoma occur in its early stages, it can be effectively mitigated through appropriate interventions [13]. Therefore, the expeditious recognition of this condition is of utmost significance. The occurrence of increased intraocular pressure is commonly linked to the pathogenesis of glaucoma, a medical condition that impacts the optic nerve within the ocular structure. The anterior chamber of the eye is comprised of a transparent fluid known as aqueous humor. This fluid plays a crucial role in regulating intraocular pressure by diligently replenishing itself on the ocular surface in a continuous and systematic manner [14]. Due to the increased pressure, the fluid is unable to demonstrate outward flow from the ocular region. Moreover, the progressive nature of the ailment results in a gradual decline of the optic nerve, which functions as the conduit for transmitting visual stimuli from the eye to the brain. As a consequence of its initial limited impact on visual acuity, glaucoma often goes unreported by individuals [15].

Ophthalmologists diagnose glaucoma using various methods. These methods include intraocular fluid volume measurement, drainage angle or outflow pathway examination, and retinal fundus examination. Since these factors are subjective and people can make mistakes, especially when comparing observers, skilled doctors are needed [10], but experienced medical professionals are limited in their availability. Additionally, the diagnostic process takes 20–45 minutes. This infection emphasizes the importance of routine eye exams for glaucoma screening. Ocular structure examinations are laborious and require skilled healthcare professionals. Retinal photography is used by ophthalmologists to diagnose glaucoma instead of manual screening. To ensure the most accurate diagnosis, images must be evaluated by highly skilled ophthalmologists [16]. At the same time, ophthalmologists often need to examine a large number of unproblematic images, which makes the professionals overburdened, so an automated solution using a machine like a computer is extremely needed. Also, current glaucoma diagnosis methods require expensive and complex equipment and specialized expertise. Thus, diagnosing many patients becomes

complicated, expensive, and time-consuming. Thus, considering an alternative approach to solving the current problem is crucial [17]. As suggested, a CA-CDSS is necessary to help ophthalmologists evaluate medical images. In the given problem, "classification" means methodically categorizing images into "abnormal" and "normal."

This study aims to explore the utilization of CA-CDSS in the context of disease classification. The research employs an optimization approach and leverages ML methodologies to achieve its objectives. The concept of computer-aided diagnosis (CAD) was first introduced in 1966 and has since been seamlessly integrated into professional practice as of 1980. CAD systems have unequivocally exhibited their inherent capability to effectively assist healthcare practitioners in making well-informed decisions with a commendable degree of certainty [18]. These decisions encompass a multitude of facets, including triage, quantification, and the diligent monitoring of promising medical cases. Furthermore, these entities possess additional advantages, such as the ability to consistently perform identical actions and introduce imperceptible modifications. The present study places significant emphasis on the utilization of CAD systems in the classification of retinal images, with the primary objective of facilitating the diagnosis of glaucoma. The proposed methodology entails the integration of bio-inspired computing techniques to facilitate FS, alongside the utilization of ML models for the purpose of classification. The retinal images utilized in this study have been acquired from benchmark datasets. In a broad sense, the procedure has three main steps: extracting features from subject images, finding the most imperative features, and then classifying the images to tell the difference between those with glaucoma and those without. As a result, the implementation of a CAD tool greatly enhances the efficiency of ophthalmologists in conducting extensive glaucoma screenings.

A successful CA-CDSS implementation for glaucoma detection depends on FS's meticulousness. These methods remove unneeded features to minimize feature space dimensionality. FS is the process of finding the smallest subset of features that can recreate the original set. Implementing FS methods generates a feature set with high discrimination. By using FS approaches, computer-based diagnostics may overcome misclassification concerns. To get the best training results, ML algorithms often require large datasets with many examples and various attributes. Alternatively, glaucoma traits may show mild oscillations, substantial relationships, or duplications. Thus, before classification, FS approaches are used to minimize the dimensionality of the feature space and collect a subset of robustly discriminating features. FS has become a key strategy for addressing extraneous features in large real-world datasets in ML. Adding irrelevant features can reduce ML system efficiency and accuracy. FS approaches aim to reduce characteristics to improve computation time and classification efficiency.

Preprocessing and categorization are fundamental initial stages in the utilization of medical images for the purpose of diagnosing glaucoma. To start, the process of making the input image less bright will make it easier to analyze it later. This will help solve problems that come up when using different filters and procedures, such as morphological operations. The initial phase of the process entails the removal of any superfluous elements or imperfections that may be present within the image. The images in the dataset underwent cropping and resizing procedures, resulting in dimensions of 224 pixels by 224 pixels. The images acquired from the dataset were converted to grayscale using the Contrast Limited Adaptive Histogram Equalization (CLAHE) technique, which improved the visual quality of the images.

The paper is organized as follows: Section 1 provides an introduction to the significance of feature selection in the context of the discussed illness as well as an overview of the computer-aided diagnostic approach employed in the study. Section 2 provides an

overview of prior studies that are pertinent to the current research. Section 3 provides a comprehensive account of the experimental design, dataset, methods, and techniques employed in the study. These include preprocessing, segmentation, feature extraction, feature selection, and classification methodologies. In Section 4, an analysis of the findings and comments is presented, encompassing performance measurements, experiments, and the selection of performance study parameters, as well as comparisons to existing approaches. Section 5 is focused on the limitations of the study and its future scope. Section 6 concludes this work.

The following is the main summary of contributions on this set of work:

1. A comprehensive set of features, including structural and non-structural indicators, is extracted from patient fundus images to classify diseases. Our current understanding is that none of the nature-inspired computing algorithms we selected have been used with this glaucoma dataset and extracted 65 features (Table 1) that could be used to find out this illness. The GWO-Machine Learning Classifier (MLC), which is based on the combination of the GWO algorithm with MLC, the WOA-MLC, which is based on the combination of the WOA algorithm with MLC, and the hGWWO-MLC, which is based on the combination of the proposed hybrid algorithm with MLC, were used to create a subset of important features and to train the MLC on the basis of this subset. This methodology not only finds the most important features but also a more precise glaucoma classifier by examining a limited set of traits. The above methods are highly efficient at detecting glaucoma, as the series of experiments yielded unmatched results.
2. In this experimental study, a thorough examination and evaluation were conducted on the ORIGA dataset, wherein a collection of six ML techniques and three soft computing techniques were utilized to analyze a benchmark dataset.
3. This study makes a significant contribution to the field of scientific research by introducing a pioneering classifier strategy that showcases quite encouraging results, achieving a maximum accuracy of 96.8% in the precise identification of glaucoma disease within the binary class. This particular model exhibits significant potential for aiding ophthalmologists in the process of diagnosing and treating a diverse range of illnesses. Furthermore, it has the potential to function as an alternative vantage point or standpoint.

Table 1 List and category of extracted features

S. No.	Type
1.	5 Structural Features (CDR, disc damage likelihood scale (DDLS), RIM Width, Cup Diameter, Disk Diameter)
2.	5 GLRM Features (Short run emphasis, Long run emphasis, Grey level run emphasis, Run percentage, Run length non-uniformity)
3.	19 GLCM Features (Autocorrelation, Contrast, Correlation, Cluster prominence, Cluster Shade, Dissimilarity, Energy, Entropy, Homogeneity, Maximum Probability, Sum of squares variance, Sum variance, Sum entropy, Difference variance, Difference entropy, Information measure of correlation2, Inverse Difference normalized, Inverse difference moment normalized)
4.	6 FOS Features (Entropy, Mean, Standard Deviation, Variance, Kurtos, Skewness)
5.	30 Wavelet Features

4. This suggested automated system employs quantitative analysis of fundus images to aid ophthalmologists in efficiently and economically identifying cases of glaucoma. The successful implementation of the CA-CDSS has given people a reliable, cost-effective, streamlined, strong, and easy-to-deploy intelligence system that was made to screen for this specific infection. As a result, it is possible that the prevalence of visual impairment could potentially be diminished.

2 Related works

This comprehensive survey is divided into two sub-sections. In the initial phase, studies that are specifically dedicated to the application of deep learning are presented. In the subsequent phase, we have ultimately chosen and thoroughly deliberated upon those studies that have employed feature engineering and ML methodologies that were not included in the initial subsection.

2.1 Deep learning based approaches for glaucoma prediction

The adoption of deep learning (DL) techniques has witnessed a significant increase in various biological imaging applications due to their remarkable capacity to generate highly efficient discriminative representations [19, 20]. The inaugural assessment of glaucoma through fundus imaging was conducted as part of the ORIGA initiative in Orlando et al. [21]. A comprehensive classification framework has been devised to facilitate the systematic categorization of glaucoma. Additionally, an evaluation methodology has been established to ensure precise segmentation of the optic disc and cup. The analysis and discussion of the twelve deep learning methodologies presented a significant challenge. The imaging modality known as colour fundus photography (CFP) has been recognized as the most economically efficient technique for the screening of retinal diseases. An AUC of 0.9885 was achieved. Bajwa et al. [22] elucidates the methodology employed by a team involved in competitive endeavors and offers a comprehensive assessment of the findings derived from their research. The assessment of the aforementioned technique was carried out utilizing the ORIGA dataset, with the primary aim of accurately categorizing cases of glaucoma [22]. The researchers successfully attained a commendable Area under the Receiver Operating Characteristic Curve (AUC-ROC) value of 0.874 in the classification of glaucoma.

The study by Guo et al. [23] elucidates the development of a smartphone application named Yanbao, meticulously engineered to optimize user-friendly accessibility and seamless navigation. The principal aim behind the development of the Yanbao App was to enhance the accessibility and convenience of glaucoma screening services for its users. The achievement of this objective is accomplished by implementing a recommended glaucoma screening algorithm that relies on clinical characteristics. A sensitivity of 0.799, specificity of 0.738, accuracy of 0.769 and AUV of 0.831 was yielded. The accurate demarcation of the optic disc and cup from fundus images holds significant importance in the screening and diagnosis of glaucoma, wherein the cup-to-disc ratio (CDR) assumes a crucial role [24–27]. The researchers in the aforementioned studies employed deep learning methodologies to effectively corroborate the presence of glaucoma. The aforementioned objective

was accomplished through the implementation of techniques including optic cup and optic disk segmentation, in addition to the quantification of the cup-to-disc ratio (CDR).

AlAfandy et al. [28] offers two effective remote sensing image classification methods that use two classic convolution models, ResNet 50 and DenseNet 169, to concatenate two convolution channel outputs as features extraction. The fully connected neural network and support vector machine classifiers employed these elicited attributes as input features. Learning curves displayed during the training of the fully connected neural network-based classifier are used to evaluate its accuracy. The suggested classifiers and their training weights predict the classes of a large remote sensing scene for a developed test. Remote sensing creates land-use/land-cover maps. Remote sensing image classification is critical (AlAfandy et al. [29]). Optimizing neural networks reduces loss functions and eliminates redundancy in the training set, improving accuracy. Genetic algorithms (GA) are most commonly employed to optimize fully connected neural networks. This study reviews deep learning techniques, including artificial neural networks, GA, and convolution neural networks. AlAfandy et al. [30] research compares evolutionary algorithms and convolution neural networks (CNNs). This comparison uses overall accuracy and kappa coefficient. This comparison indicates that various factors affect classifier accuracy. CNNs are more accurate than GAs and cost less to compute. This research compares deep traditional convolution networks for remote sensing picture categorization [30]. This comparison analysis uses DenseNet196, NASNet Mobile, VGG16, and ResNet50 deep convolution models. These learning convolution models employ ImageNet pretrained weights, transfer learning, and a complete connected layer that matches the dataset classes. This comparison uses the SIRI-WHU and UC Merced land use datasets. The analysis shows that the ResNet50 model has the best accuracy and the NASNet Mobile model the lowest. DenseNet169 is somewhat more accurate than VGG 16.

2.2 Feature selection and machine learning methods in prediction of glaucoma

A wide array of methodologies has been employed in the classification of glaucoma, encompassing a variety of features extracted from retinal images. The research conducted by Rajendra et al. [31] presented a computational methodology for the diagnosis of glaucoma, employing high order spectra (HOS) and textural characteristics. The study utilized Support Vector Machine (SVM), Naive Bayes (NB), and Random Forest (RF) algorithms for the purpose of the categorization task [19] and achieved the accuracy of more than 91%. Subsequently, Sumeet et al. [32] employed wavelet features to classify glaucoma using SVM, Self-Organizing (SO) maps, NB, and RF algorithms. The research findings indicate that SVM exhibited a significantly higher level of performance in comparison to alternative classifiers with an accuracy rate of 93% [32]. Furthermore, the study conducted by Mookiah et al. [33] employed SVM as a method to effectively detect glaucoma by utilizing wavelet and Higher Order Statistics (HOS) features; an accuracy of 95%, sensitivity and specificity of 93.33% and 96.67% respectively was gained. In their study, Noronha et al. [34] employed a SVM and NB classifier to extract HOS and conduct glaucoma classification. The NB classifier demonstrated superior performance when compared to the SVM, achieving an accuracy rate of 92.65%, sensitivity of 100% and specificity of 92%. In a subsequent study, Acharya et al., [35] employed Gabor features in conjunction with SVM and NB classifiers to effectively classify cases of glaucoma. SVM gives the highest average accuracy of 93.10%, sensitivity of 89.75% and specificity of 96.20% using 23 features. Moreover, Isaac et al. [36] performed an extraction of structural attributes from

a retinal image that underwent segmentation in order to isolate the disc and cup regions. The aforementioned attributes encompassed the CDR and the disc damage likelihood scale (DDLS). Subsequently, the process of classification was executed utilizing SVM and a neural network (NN) [36]. An accuracy of 94.11% with sensitivity of 100% is achieved using SVM. In a subsequent investigation, Salam et al. [37] employed a SVM classifier to extract features pertaining to structure, texture, and intensity. The average sensitivity and specificity of proposed system are 100 and 87 % respectively. [37]. Haleem et al.[38] introduced a methodology for the classification of abnormal and normal glaucoma utilizing a SVM classifier with classification accuracy rate of 94.4%. The study employed a diverse set of classification characteristics, encompassing structural, Gaussian, Gabor, and wavelet methods. Furthermore, Claro et al.[39] undertook a comprehensive study wherein they systematically classified retinal images utilizing various classification methodologies, including maximum linear progression (MLP), RF, and radial basis function (RBF). The researchers gathered a diverse range of structural and textural attributes, which were subsequently evaluated using multiple classifiers. The MLP classifier demonstrated superior performance when compared to other classifiers, achieving an accuracy rate of 93.03% [39]. Singh et al. [40] proposed an automatic image analysis system to diagnose glaucoma using wavelet features from the segmented optic disc in digital fundus images. The noisy blood vessels were removed from the segmented optical disc, and features were extracted from the de-noised image. Glaucoma images were classified with 94.7% accuracy using first-level wavelet features from segmented and blood vessel-in-painted optic disc images, outperforming related works. Five supervised classifiers were tested for glaucoma image classification using evolutionary attribute selection and principal component analysis to select prominent features. All five classifiers have accuracy over 85%. Random forest and artificial neural network classifiers can detect glaucoma with 94.7% accuracy using evolutionary attribute selection, while SVM and k-NN use principal component analysis.

Sousa et al. [41] employed SVM as a methodology for extracting geostatistical functions predicated on texture. The aforementioned functions encompass the semivariogram, semivariogram, correlogram, and covariogram. The SVM classifier achieved an accuracy rate of 91.2%, a sensitivity rate of 95%, and a specificity rate of 88.23% [41]. Subsequently, Koh et al. [42] employed a random forest classifier to extract wavelet-based features from retinal fundus images. The study has yielded findings indicating accuracy, sensitivity, and specificity values of 92.4%, 89.3%, and 95.5% respectively. In a study conducted by Septiarini et al. [43], a comprehensive analysis was carried out on retinal images. The authors directed their attention towards the extraction of statistical characteristics from a designated region of interest (ROI). The aforementioned characteristics were subsequently classified using the k-Nearest Neighbors (k-NN) algorithm, resulting in a precision rate of 95.24%. In addition, Selvathi et al. [44] conducted an extraction of wavelet-based features with the objective of utilizing a neural network for classification purposes. The outcome of this study revealed an accuracy rate of 95.8%, a specificity rate of 91.66%, and a sensitivity rate of 100% [44]. In a recent study conducted by Shubhangi et al. [45], the researchers investigated the characteristics of cross entropy in wavelet processed input fundus images. The aforementioned characteristics were subsequently assessed and ranked using statistical tests. The subsequent classification task was executed utilizing SVM with enhanced efficacy. . The classification efficiency in classifying the normal and glaucoma affected image is found to be 99%. [45]. Moreover, Renukalatha [46] employed a comprehensive classification methodology for glaucoma that synergistically incorporated structural factors, including the cup-to-disc ratio (CDR), and non-structural parameters, such as texture. The aforementioned characteristics were subsequently classified with a 94% accuracy utilizing a multi-level SVM.

In recent years, there has been a notable increase in the scholarly literature that extensively documents the widespread adoption of automated techniques for the diagnosis of glaucoma through the meticulous analysis of fundus images. Based on the research conducted by Bock et al. [47], the initial phase of image preprocessing encompassed the manipulation of lighting conditions and the removal of blood vessels. Subsequently, the procedure of image normalization was undertaken. Subsequently, the process of extracting features was conducted utilizing B-spline coefficients, Fourier coefficients, and raw intensity values. Ultimately, a classifier was developed through the utilization of a SVM algorithm. An AUC of 87% was attained. In their research, Acharya et al.[35] employed the Gabor transform technique for the purpose of image manipulation. The utilization of Gabor coefficients facilitates the extraction of a diverse array of statistical measures, encompassing, yet not restricted to, mean, variance, skewness, energy, kurtosis, Shannon entropy, Renyi entropy, and Kapur entropy. The aforementioned measures were subsequently employed in the classification procedure by utilizing SVM. An average accuracy of 93.10%, sensitivity of 89.75% and specificity of 96.20% was gained. In a study conducted by Raghavendra et al. [48], the researchers obtained segmented optic disc images and subsequently employed the Radon transform technique. The implementation of the Modified Census Transform (MCT) was employed to address the impact of illumination variations. Subsequently, the MCT image is subjected to the application of the GIST descriptor, aiming to extract the energy spectrum of the spatial envelope. Following the aforementioned process, the acquired data is subjected to classification utilizing the SVM. Their system achieved maximum accuracy (93.62%), sensitivity (87.50%) and specificity (98.43%) using *twenty six* features. In their research, Maheshwari et al.[49] employed the 2D-variational mode decomposition (VMD) technique to accomplish image decomposition. The features employed in this study encompass Kapur entropy, Renyi entropy, and Yager entropy. The method achieved classification accuracies of 95.19% and 94.79% using three-fold and ten-fold cross-validation strategies, respectively using SVM. In their study, Raja et al.[50] employed various image decomposition techniques, including complex wavelet transform (CWT) [50], wavelet packet decomposition (WPD) [51], and optimum hyper analytic wavelet transform (OHAWT) [52]. Moreover, a SVM was utilized for the purpose of classification. Proposed technique has achieved accuracy value of 81% using SVM. In their study, Maheshwari et al. [53] employed a 2D-empirical wavelet transform (2D-EWT) technique for the purpose of converting the RGB image. This process yielded the generation of sub-images. The incorporation of correntropy as a feature was executed in tandem with the least-squares support vector machine (LS-SVM) classifier. The classification accuracy of 98.33% and 96.67% was gained using threefold and tenfold cross validation, respectively.

The classifier developed by Karsu et al. [54] integrates features derived from CDR and anisotropic dual-tree complex wavelet transform. The classifier was integrated with a multi-layer perception (MLP) network. The results show that the method achieved an accuracy rate of 97.67% with 98%. In their research, Sharma et al. [55] employed a SVM classifier to effectively classify cases of glaucoma. The aforementioned classification was derived from a comprehensive analysis of 14 statistical features, meticulously extracted from high order spectra. The proposed algorithm has yielded classification accuracy of 98.8% and 95% using entire private and public databases, respectively. The research conducted by Maheshwari et al. [56] employed a methodology that involved extracting statistical characteristics using local binary patterns from each individual bit-plane of every channel in RGB images. The aforementioned characteristics were subsequently utilized to generate statistical features, which were then employed in a classification process utilizing a SVM classifier. An accuracy of 99.30% using 10-fold cross validation was achieved. The

researchers, Agrawal et al. [57], employed the quasi-bivariate Variational Mode Decomposition (QB-VMD) technique in their study to perform the decomposition of fundus images. The LS-SVM classifier is utilized for the purpose of classifying a collection of 70 features that have been extracted from deconstructed images in a thorough manner. The 85.94 and 86.13% accuracies were acquired using three- and ten-fold cross validation, respectively.

In a previous study Jerith & Kumar[58], researchers utilized a hybrid approach known as Grey Wolf Optimized with Neural Network (GWO-NN) in order to facilitate the detection of glaucoma. However, it is worth mentioning that a limitation was identified in terms of the number of extracted features, and the authors opted not to disclose the dataset used for validating their proposed methodology. The accuracy of the proposed GWO-NN differs from NN, Naïve Bayes, and SVM by 0.06896, 0.20689 and 0.10344 respectively.

Based on a comprehensive analysis of various published literature, it can be inferred that the GWO and WOA have exhibited significant effectiveness in tackling a wide range of optimization problems, including the prediction of human diseases [59–65]. However, the application of these methodologies to augment characteristics in glaucoma classification remains somewhat constrained. The identification of this research gap serves as the primary motivation for our decision to utilize these algorithms in the validation of this infection. Based on an extensive analysis conducted on the scholarly platform scholar.google.com, it is evident that there are very limited published research studies concerning the optimization of features for the prediction of glaucoma utilizing the GWO and WOA. There is a scarcity of research that has proficiently employed these bio-inspired algorithms to optimize the selected features related to this illness. The principal objective of this study is to address the current research void and introduce a novel hybrid methodology to the scientific and medical communities, particularly targeting researchers and ophthalmologists.

3 Materials and methods

The methodology described in this study utilizes specific techniques to extract a total of 65 features from glaucoma fundus retinal pictures, as outlined in Table 1. The GWO, WOA and hGWWOA were individually applied to the set of 65 retrieved features for efficient feature selection process. In the subsequent phase, the selected features are employed to classify subject images into two categories: images of healthy individuals and images of patients with glaucoma. This classification task is facilitated by six ML algorithms functioning as classifiers. Figure 1 illustrates a diagrammatic representation of the entire work.

3.1 Preprocessing and extraction of structural features

To ascertain the cup-to-disc ratio (CDR), it is imperative to initially segment the optic disc and optic cup. In the present procedure, the extraction of the region of interest (ROI) is performed from the input fundus image. The return on investment (ROI) is allocated across three distinct channels, namely Red, Green, and Blue. The red component is deemed to be highly suitable for the purpose of optic disc segmentation due to its abundant information that is specifically relevant to the disc region. Conversely, the green component is regarded as optimal for the segmentation of the optic cup due to its inclusion of the suitable region for this anatomical structure. The primary objective of employing the CLAHE procedure is to enhance the contrast within the fundus image. The CLAHE methodology is widely employed in the

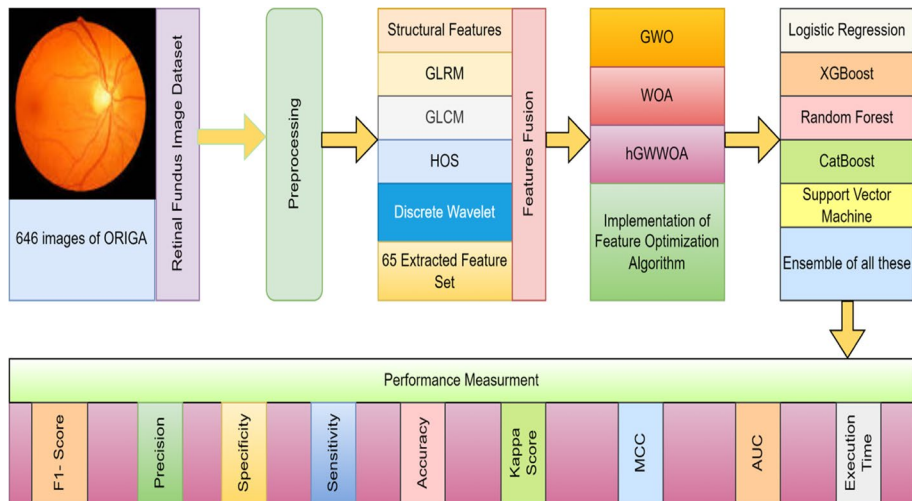


Fig. 1 Diagrammatic representation of the whole implemented work

domain of image processing to effectively enhance the contrast of photographs. The subsequent phase entails the implementation of median filtering. The filter under consideration is a nonlinear filter that exhibits exceptional suitability for the purpose of enhancing the retinal image. Based on our comprehensive methodology, it has been ascertained that the median filter stands out as the most optimal selection when considering a range of low pass filters. This phenomenon occurs due to its inherent capability to retain low-frequency elements while efficiently attenuating high-frequency elements. The improvement of output accuracy can be achieved by acquiring more precise data through pre-processing techniques specifically applied to the green channel. The process of determining the optimal threshold value for the segmentation of the optic disc and cup is regarded as a highly intricate task. To ascertain the threshold value of the optic cup, it is imperative to compute the mean of the pre-processed green channel, the standard deviation of the green channels, and the standard deviation of the Gaussian filtered image using a filter size of 60. The subsequent equation is employed for the computation of the threshold value. A sequence of morphological operations was conducted on the magnified image in order to extract the blood vessels. The successful extraction of the optic disc and optic cup area from the retinal fundus image has been accomplished (Eq. 1). The flow sequence employed in the preprocessing stage is depicted in Fig. 2.

$$\text{Thg} = 0.5 * \text{Mg} + 2 * \text{STDf} + 2 * \text{SDg} + \text{Mg} \quad (1)$$

The variable "Mg" is used to indicate the mean value of the green channel, whilst "STDf" is utilized to express the standard deviation of the filter. Furthermore, the term "SDg" represents the standard deviation of the green channels, whereas "Mg" once again represents the mean value of the green channels. The variable "THG" denotes the threshold value for the optic cup. Upon the application of the acquired threshold value on the green channel, we have successfully segmented the region corresponding to the optic cup.

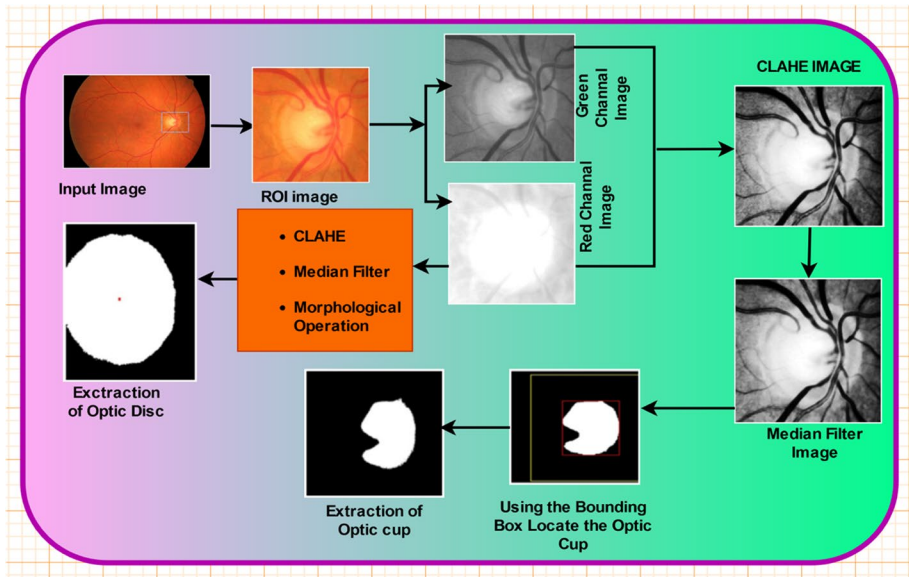


Fig. 2 Sample for extraction optic cup and optic disc from Retinal fundus image

In order to forecast the threshold value of the optic disc, it is necessary to compute the average and standard deviation of the pre-processed red channels. The subsequent formula (2) is employed to derive the threshold value for the optic disc.

$$\text{Thr} = 0.5 * M - \text{STDf} - \text{Ar.std} () \quad (2)$$

Here Thr is the threshold value for optic disc. After applying this threshold value on red channels we have segmented optic disc.

The subsequent objective involves the calculation of the CDR utilizing the segmented regions of the cup and disc. To optimize the visual fidelity and accurately delineate the periphery of the optic disc, we employ supplementary morphological operations, namely opening and closing operations, on the optic cup. In the subsequent phase, we will proceed with the computation of all viable outlines. After careful evaluation, we have conducted an assessment of the various contours and determined that the contour exhibiting the greatest area has been identified. The subsequent procedure entails the determination of the major axis of the ellipse that possesses the utmost area. In order to accomplish this task, the fit ellipse technique is utilized, with particular emphasis placed on identifying the ellipse possessing the greatest area. Subsequently, the procedure of aligning a rectangle onto the ellipse is utilized to ascertain the magnitude of the major axis. The dimension denoting the greatest distance across an ellipse, commonly known as the major axis, is occasionally referred to as the cup diameter. In a comparable manner, the computation of the optic disc's diameter has been conducted. The CDR value is derived through the division of the cup diameter by the disc diameter. The Disc-to-Disc Diameter (DDLS) is a metric employed to assess the thickness of the ocular surface relative to the disc diameter, with the objective of reducing the incidence of glaucomatous pathology. This methodology quantifies the degree of damage caused by

glaucoma, evaluates the rate at which the disease is progressing, and monitors its severity or progression. In the event that the ratio surpasses 0.6, it can be deduced that the eye manifests indications of glaucoma. Conversely, if the ratio falls below 0.6, the eye is categorized as being within the normal range.

3.2 Features optimization algorithm

3.2.1 Grey Wolf Optimization (GWO)

The GWO algorithm comes under the category of Swarm Intelligence Algorithms and is one of the Nature-inspired Algorithms [66]. It was primarily inspired by grey wolves' natural social leadership and hunting behaviour. The best solutions are alpha (α), beta (β), and delta (δ) according to the algorithm. They are the three wolves in charge. They are in charge of leading the remainder of the wolves, known as omega (ω) wolves, to the global solution [67].

The algorithm comprises of very few parameters and it is very easy to implement. GWO has a proposed approach, as it is based on communal conduct of grey wolves and their leadership hierarchy. Grey wolves usually live in packs of 5 to 12 wolves. The group leader is the alpha wolf. They make all the major decision for the group. Therefore, alpha wolves are known as dominant wolves, they command other wolves to obey their order. The beta wolves, who are second in command after the alpha wolf, are the next in line. They follow and listen to alpha wolf orders. They take command in case the alpha wolf dies. The deltas are the subordinate wolves. They can have various posts in the group such as elders, scouts, caretakers, hunters, sentinels. The alpha and beta wolves generally instruct the delta wolves and they also handle the omega wolves. Finally, the omega wolves they are very low rank wolves, they generally play the part of scapegoat. They are obliged to follow all the instructions from the alpha, beta and delta wolves. Not being significant wolves, they help in solving internal problems. Table 2 depicts the parameter settings for GWO.

The Optimizer consists of mainly three steps -:

1. Encircling
2. Hunting
3. Attacking

Table 2 Parameter Settings GWO

Parameters	Values
No. of Iterations	100, 200, 300, 400, 500
Population Size	5 ,10 ,15 ,20
Number of Features	65
Number of Instances	646
LB (lower bound)	-100
UB (upper bound)	100
Control Parameter (ψ)	Linearly decreases from 2 to 0
Objective Function	Rastrigin Function

Encircling prey The grey wolves first encircle the prey which is modeled by these equations given below,

$$M = |N\eta_p(l) - \eta(l)| \quad (3)$$

$$\eta(l+1) = \eta p(l) - \kappa \cdot M \quad (4)$$

Here η_p is the prey position, η indicates the position vector of grey wolf, and l is the current iteration. κ and N are coefficient vectors.

$$\kappa = 2 \cdot \psi(l) \cdot \text{ran}1 - \psi(l) \quad (5)$$

$$N = 2 \cdot \text{ran}2 \quad (6)$$

Here $\text{ran}1$ and $\text{ran}2$ are random vectors between $[0,1]$, and value of vector ψ will linearly decrease from 2 to 0 in upcoming iterations.

$$\psi(l) = 2 - (2 \cdot l) / \text{MaxIter} \quad (7)$$

Hunting Because alpha, beta, and delta are the best hunters, it is considered that they have a better understanding of where prey can be found. As a result, the surviving omega wolves must trail the best wolves. The following equation will describe the hunting behavior of wolves:-

$$\begin{aligned} M\alpha &= |N1 \cdot \eta\alpha - \eta(l)| \\ M\beta &= |N2 \cdot \eta\beta - \eta(l)| \\ M\delta &= |N3 \cdot \eta\delta - \eta(l)| \end{aligned} \quad (8)$$

Here $N1$, $N2$ and $N3$ are calculated by Eq. 6

$$\begin{aligned} \eta i1(l) &= \eta\alpha(l) - \kappa i1 \cdot M\alpha(l) \\ \eta i2(l) &= \eta\beta(l) - \kappa i2 \cdot M\beta(l) \\ \eta i3(l) &= \eta\delta(l) - \kappa i3 \cdot M\delta(l) \end{aligned} \quad (9)$$

Here $\eta\alpha$, $\eta\beta$ and $\eta\delta$ are the first three best solutions at iteration l , $\kappa i1$, $\kappa i2$ and $\kappa i3$ is calculated in Eq. (5) and $M\alpha$, $M\beta$ and $M\delta$ are defined in Eq. (6).

$$\eta(l+1) = \frac{\eta i1(l) + \eta i2(l) + \eta i3(l)}{3} \quad (10)$$

Attacking When the prey stops moving, the hunting process comes to an end, and the wolves begin to attack. This is done as the value of ψ will start to decrease in the upcoming iterations controlling the exploration and exploitation of the prey by grey wolves. In each iteration, Eq. (7) is changed from range 2 to 0. Approximately half of the iterations are dedicated to exploration, with a smooth transition, and the other half to exploitation. Wolves shift their positions between the prey position and their current position at random. This technique works by producing an initial population of wolves within the search space at random. The wolves' position is evaluated using the objective function. The stages are continued until the stopping condition, or the predetermined number of iterations, is met. The first three wolves with the best fitnesses are considered in each iteration as α , β , and δ . The wolves then update their positions in relation to the previous processes of circling, hunting, and attacking. As a result, the best prey location, i.e. the alpha's position, can be determined.

Pseudocode

Input: Population Size, Dimension, Maximum Iterations, Objective Function

Output: The Global Optimum

Step 1: Begin
 Step 2: Getting started (Randomly distributing N wolves in the search space and calculating their fitness).
 Step 3: While iter = 1 to MaxIter
 Step 4: Find $\eta\alpha$, $\eta\beta$ and $\eta\delta$
 Step 5: For i = 1 to N
 Step 6: Computing $\eta i1$, $\eta i2$, $\eta i3$ by using Eq. (9).
 Step 7: Computing $\eta i(l+1)$ by using Eq. (10).
 Step 8: Selecting best ($\eta i(l+1)$)
 Step 9: Updating Pop.
 Step 10: End for
 Step 11: End while
 Step 12: Return the global optimum.
 Step 13: End

Variables used in the algorithm:

ηp : Prey position η : Grey wolf's position vector
 κ , N : Coefficient vectors ψ : decreased linearly from 2 to 0
 M : Distance calculated between Prey and Grey wolf; $ran1$, $ran2$: Random vector between [0,1]

Objective function The objective Function used for the GWO, WOA and their hybrid (hGWAO) is same which is Rastrigin Function. Mathematical implementation of Rastrigin Function-:

$$f(x) = \sum_{i=1}^n [xi^2 - 10 \cos(2\pi xi) + 10]$$

The Rastrigin function is mainly used as a performance test problem for optimization algorithms. It is basically a non-linear multimodal function was first proposed in 1974. The function has many local minima it is because the optimization algorithms have to be run from multiple starting points. This function has only one global minimum, and all other local minimums have value greater than global minimum. It has been studied that GWO shows premature convergence and can be stuck in local minima, therefore multimodal functions can test the ability of the GWO algorithm in reaching the global minima by avoiding local minima.

Flowchart The flow chart of the GWO is depicted using Fig. 3

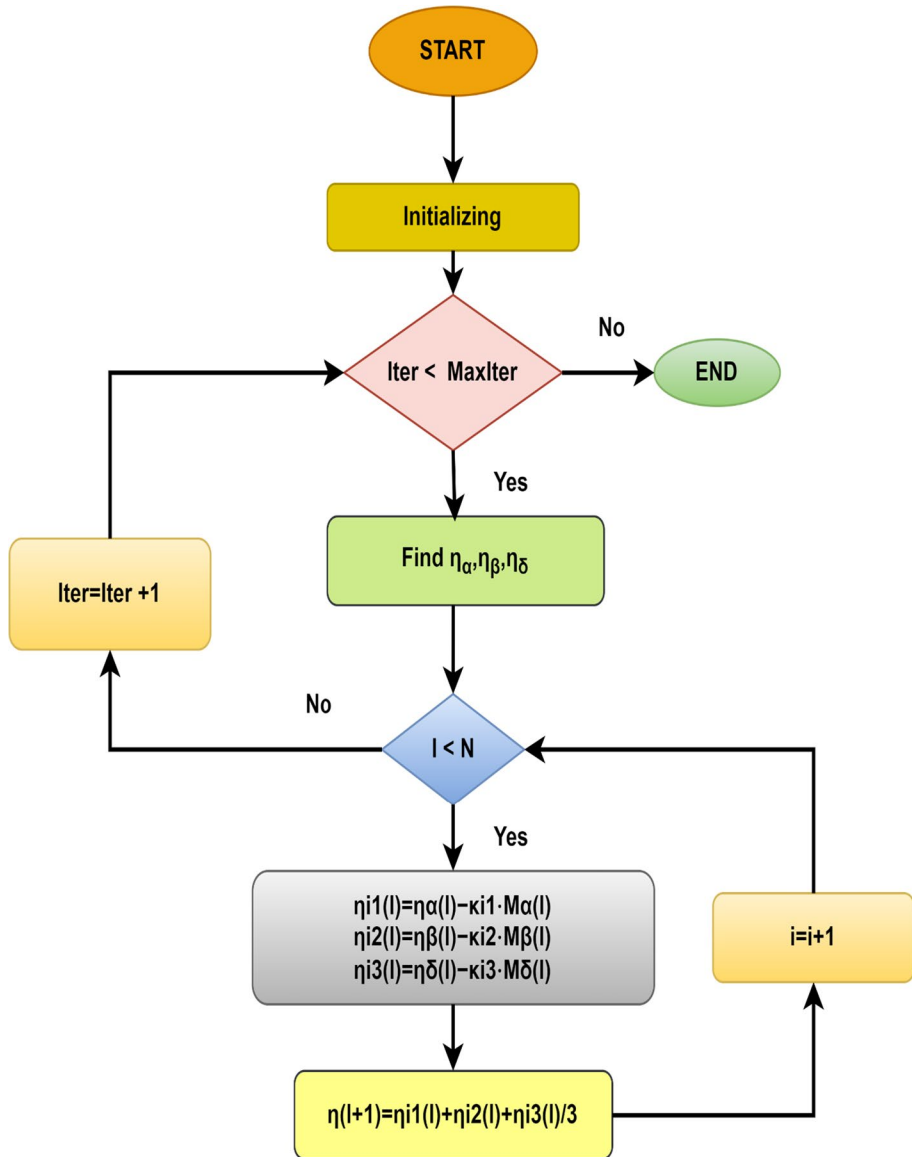


Fig. 3 Flowchart of the GWO algorithm

3.2.2 Whale Optimization Algorithm (WOA)

Whales are among the world's largest mammals. Whales can reach 30 metres in length and weigh 180 tonnes. Sei, Killer, Minke, Humpback, Blue, Finback, and Right whales are among the various whale species found on the earth.

They are unable to sleep since they must breathe from the ocean's surface. Whales are said to be extremely intelligent creatures with feelings. Whales have a few common cells that are similar to human spindle cells. Emotions, social behaviour, and judgement are all

controlled by these cells in humans. Whales have twice the number of cells as an adult person, which explains their intelligence. Certain whales may establish their own dialects, and it has been demonstrated that whales can learn, judge, communicate, and even feel emotionally affected in the same way as humans can.

The WOA is based on humpback whale hunting behavior [68]. According to a study in the social behaviour of Whales it has been proven that whales can live alone or in groups but mainly the whales are observed in groups [68]. Killer whales live with an entire group throughout their lifetime. The size of an adult humpback whale is comparable to that of a school bus. Small fish herds or krills are their preferred prey. The humpback whale's hunting method is one of the things that fascinate us about them [69]. They mostly employ the bubble net attack. Humpback whales tend to hunt krill or small fish that are found near the surface. The hunt is carried out by blowing bubbles in a 9-shaped trail around a circle. Humpback whales dive 12 meters down to their meal and begin to form spiraling bubbles before capturing their prey while plunging upwards [69]. The bubble net feeding technique is unique to humpback whales and can only be seen in them. For optimization, the spiral bubble net feeding manoeuvre is mathematically modeled.

Encircling prey Humpback whales are natural predators that can locate their prey and encircle it once found. The algorithm assumes the best candidate solution as their target prey because the ideal location of the prey is unknown a priori. When the best search agent is identified, the remaining search agents will attempt to change their locations in order to align with the best search agent.

Hence, we can represent this from the equations given below -:

$$M = |N \cdot \eta^*(t) - \eta(t)| \quad (11)$$

$$\eta(t+1) = \eta^*(t) - \kappa \cdot M \quad (12)$$

Here t indicates the current iteration, κ and N are coefficient vectors, η^* is the position vector of the best solution, and η is the position vector. η^* will be updated at each iteration if there exists a better solution.

We can calculate the coefficients from these equations-:

$$\kappa = 2 \cdot \iota - \partial \quad (13)$$

$$N = 2 \cdot \iota \quad (14)$$

Here ∂ will be decreased from 2 to 0 linearly in the upcoming iterations (in both exploration and exploitation phase) here we define ι as a random vector between [0,1]. By modifying the value of, we can get to several areas surrounding the optimal answer in relation to our current position of κ and N . As a result, every search agent can update its position in the neighbourhood and simulate encircling prey using Eq. 12. As a result, we may apply the same notion to an n -dimensional search space, with search agents moving in hyper-cubes around the optimal solution.

Bubble-net attacking method (exploitation phase) The bubble-net attacking method now has two approaches.

1. Shrinking Encircling Mechanism:

This behaviour is achieved by decreasing value of ∂ in Eq. 11. The fluctuation range of κ is also decreased by ∂ . Hence κ will lie randomly in the interval $[-\partial, \partial]$ where ∂ is decreased from 2 to 0.

2. Spiral Updating Position:

The distance between the whale and the prey is calculated using this method. The position of the whale and the prey are then used to generate a spiral equation, which results in a helix-shaped movement.

$$\eta(t+1) = M \cdot e^{bl} \cdot \cos(2\pi l) + \eta^*(t) \quad (15)$$

Where $M' = |\eta^*(t) - \eta(t)|$ and indicates distance of the i^{th} whale to prey, b is a constant for defining shape of logarithmic spiral and l is random number in $[-1, 1]$.

To update the position of whales, there is now a 50% chance of choosing either shrinking encircling or the spiral model.

$$\begin{aligned} \eta(t+1) &= \eta^*(t) - \kappa \cdot M, & \text{If } p < 0.5 \\ \eta(t+1) &= M' \cdot e^{bl} \cdot \cos(2\pi l) + \eta^*(t), & \text{If } p \geq 0.5 \end{aligned} \quad (16)$$

Where p is random number between $[0, 1]$. The humpback whales also search for prey randomly

Search for prey (exploration phase) Based on the variation of the κ vector can be utilized to look for prey (exploration). Humpback whales search at random, based on their relative positions. Hence, we use κ with the random values greater than 1 or less than -1 to force search agent to move far away from a reference whale. In the exploration phase, we update the position of a search agent using a randomly picked search agent rather than the best search agent identified. This mechanism and $|\kappa| > 1$ emphasize exploration and allow the WOA algorithm to perform a global search. The mathematical model is:

$$M = |N \cdot \eta_{rand} - \eta| \quad (17)$$

$$\eta(t+1) = \eta_{rand} - \kappa \times M \quad (18)$$

where η_{rand} is a random position vector (a random whale) chosen from the current population. WOA includes only two main internal parameters to be adjusted (κ and N). Parameter settings for WOA are shown using Tables 3 and 4.

Pseudocode

Table 3 Parameter Settings WOA

Parameters	Values
No. of Iterations	100, 200, 300, 400, 500
Population Size	5, 10, 15, 20
Number of Features	65
Number of Instances	646
LB (lower bound)	-100
UB (upper bound)	100
Constant for smoothening shape of logarithmic spiral (b)	1
Objective Function	Rastrigin Function

Table 4 Parameter Settings hGWWOA

Parameters	Values
No. of Iterations	100, 200, 300, 400, 500
Population Size	5 ,10 ,15 ,20
Number of Features	65
Number of Instances	646
LB (lower bound)	-100
UB (upper bound)	100
Control Parameter (ψ)	Linearly decreases from 2 to 0
Constant for defining shape of logarithmic spiral (χ)	1
Objective Function	Rastrigin Function

Step 1: Initialize the whales population η_i ($i=1, 2, \dots, n$)
 Step 2: Calculate each search agent's fitness.
 Step 3: η^* = the best search agent
 Step 4: while ($t < \text{maximum number of iterations}$)
 Step 5: for each search agent
 Step 6: Update ∂ , κ , N , l and p
 Step 7: If ($p < 0.5$)
 Step 8: If ($|k| < 1$)
 Step 9: Eq. (11) updates the current search agent's position
 Step 10: else if ($|k| \geq 1$)
 Step 11: Select a random search agent (η_{rand})
 Step 12: Update the position of the current search agent by Eq. (18)
 Step 13: else if ($p \geq 0.5$)
 Step 14: Update position of current search agent by Eq. (14)
 Step 15: end if
 Step 16: end for
 Step 17: Check to see if any search agents reach beyond the search space, and if so, make the necessary changes.
 Step 18: Update η^* if there is a better solution
 Step 19: $t = t + 1$
 Step 20: end while
 Step 21: return η^*

Variables used in the Algorithm

κ , N : coefficient vectors ; η^* : The optimal solution's position vector
 η : Position vector. ; ∂ : linearly decreased from 2 to 0
 M : Indicates distance of the i th whale to prey ; b : Constant for defining shape of logarithmic spiral
 l : Random number between $[-1, 1]$; p : Random number between $[0, 1]$
 r : Random vector between $[0, 1]$; t : Used for iterations

Flowchart Figure 4 depicts the flowchart of the WOA algorithm.



Fig. 4 Flowchart of the WOA algorithm

3.2.3 GWO-WOA Hybrid Algorithm (hGWWOA)

hGWWOA is predicted to increase the WOA Algorithm's performance. The given method applies the GWO leadership ranking to the WOA's bubble net attacking strategy. The proposed algorithm selects three best candidate solutions from the entire search agents, the first level α , the second and third levels in the group β and δ , and the other search agents will adjust their positions in accordance with the positions of the best search agents to improve the performance of the WOA algorithm.

Humpback whales use two ways to swim around their prey, as previously indicated. The mathematical model that has been proposed for updating whale positions. The following is a

mathematical representation of this during optimization utilizing GWO's leadership hierarchy. Bubble-net attack strategy is implemented. They use of the hierarchy to update the position of whales.

GWO equations are

$$M = |N \cdot \eta p(m) - \eta(m)| \quad (19)$$

$$\eta(m+1) = \eta p(m) - \kappa \cdot M \quad (20)$$

$$\kappa = 2 \cdot \psi(m) \cdot \text{ran1} - \psi(m) \quad (21)$$

$$N = 2 \cdot \text{ran2} \quad (22)$$

$$\begin{aligned} M\alpha &= |N1 \cdot \eta\alpha - \eta(m)| \\ M\beta &= |N2 \cdot \eta\beta - \eta(m)| \\ M\delta &= |N3 \cdot \eta\delta - \eta(m)| \end{aligned} \quad (23)$$

$$\begin{aligned} \eta i1(m) &= \eta\alpha(m) - \kappa i1 \cdot M\alpha(m) \\ \eta i2(m) &= \eta\beta(m) - \kappa i2 \cdot M\beta(m) \\ \eta i3(m) &= \eta\delta(m) - \kappa i3 \cdot M\delta(m) \end{aligned} \quad (24)$$

$$\eta(m+1) = \frac{\eta i1(m) + \eta i2(m) + \eta i3(m)}{3} \quad (25)$$

Whale equations are

$$\eta(m+1) = M' \cdot e^{bl} \cdot \cos(2\pi l) + \eta * (m) \quad (26)$$

$$M' = |\eta * (m) - \eta(m)| \quad (27)$$

$$\begin{aligned} \eta(m+1) &= \eta * (m) - \kappa \cdot M, \text{ if } p < 0.5 \\ \eta(m+1) &= M' \cdot e^{bl} \cdot \cos(2\pi l) + \eta * (m), \text{ if } p \geq 0.5 \end{aligned} \quad (28)$$

$$M = |N \cdot \eta_{rand} - \eta| \quad (29)$$

$$\eta(m+1) = \eta_{rand} - \kappa \times M \quad (30)$$

The following is a formula for GWO leadership: Humpback whales have a shrinking encircling mechanism utilizing to update their position using Eq. (22).

Mathematical Model and Algorithm for Optimization Spiral updating position: To update the position of humpback whales on a spiral path, use the formula below.

$$\begin{aligned} M'_\alpha &= |\eta_\alpha(m) - \eta(m)| \\ M'_\beta &= |\eta_\beta(m) - \eta(m)| \\ M'_\delta &= |\eta_\delta(m) - \eta(m)| \end{aligned} \quad (31)$$

$$\begin{aligned} \eta_1(m) &= \eta_\alpha(m) + M'_\alpha \cdot e^{\chi\varphi} \cdot \cos(2\pi\varphi) \\ \eta_2(m) &= \eta_\beta(m) + M'_\beta \cdot e^{\chi\varphi} \cdot \cos(2\pi\varphi) \\ \eta_3(m) &= \eta_\delta(m) + M'_\delta \cdot e^{\chi\varphi} \cdot \cos(2\pi\varphi) \end{aligned} \quad (32)$$

$$\eta(m+1) = \frac{\eta_1 + \eta_2 + \eta_3}{3} \quad (33)$$

Pseudo code The pseudo code for the proposed hGWWOA algorithm can be built in the following steps:

Step 1: Create the search agent's initial population.
 Step 2: Determine the value of the objective function for each search agent.
 Step 3: η_{α} = The most suitable candidate
 Step 4: η_{β} = The second possible solution
 Step 5: η_{δ} = The Third candidate solution
 Step 6: While ($m < \text{Max number of iterations}$)
 Step 7: for $i = 1$ to number of each search agent
 Step 8: Make a change to control parameter (κ , N , ψ , l and q).
 Step 9: If $i = (q < 0.5)$,
 Step 10: if $2 (|\kappa| < 1)$
 Step 11: Update the search agent's position by (25)
 Step 12: else If $2 (|\kappa| > 1)$
 Step 13: Pick a search agent at random (η_{rand}).
 Step 14: Update the search agent's position by (30).
 Step 15: end if₂
 Step 16: else if $1 (q \geq 0.5)$
 Step 17: Update the position of the search agent by (33).
 Step 18: end if₁
 Step 19: end for
 Step 20: Check to see whether any search agents have left the search space.
 Step 21: Calculate the value of each search agent's objective function.
 Step 22: Update the position of η_{α} , η_{β} and η_{δ} .
 Step 23: $m = m + 1$
 Step 24: end while
 Step 25: return η_{α} .

Variables used in the algorithm η_{α} : The best candidate solution ; η_{β} : The second candidate solution

η_{δ} : The Third candidate solution ; q : random number in $[0,1]$

φ : Random number in $[-1,1]$; χ : The shape of the logarithmic spiral is defined by this constant.

η : Humpback whale position vector; m : used for iterations

Flowchart Figure 5 depicts the flowchart of the WOA algorithm.

3.2.4 Time complexity of three implemented soft-computing algorithms

The time complexity of GWO is $O(NdI)$, where N is the number of grey wolves, d is the number of dimensions, and I is the maximum number of iterations. This means that the running time of GWO is proportional to the product of these three parameters.

One of the factors that affect the performance of WOA is its computational complexity, which is related to the number of operations required to execute the algorithm. The computational complexity of WOA depends on several parameters, such as the number of iterations, the number of search agents (whales), and the dimensionality of the problem. The computational complexity of WOA can be estimated as:

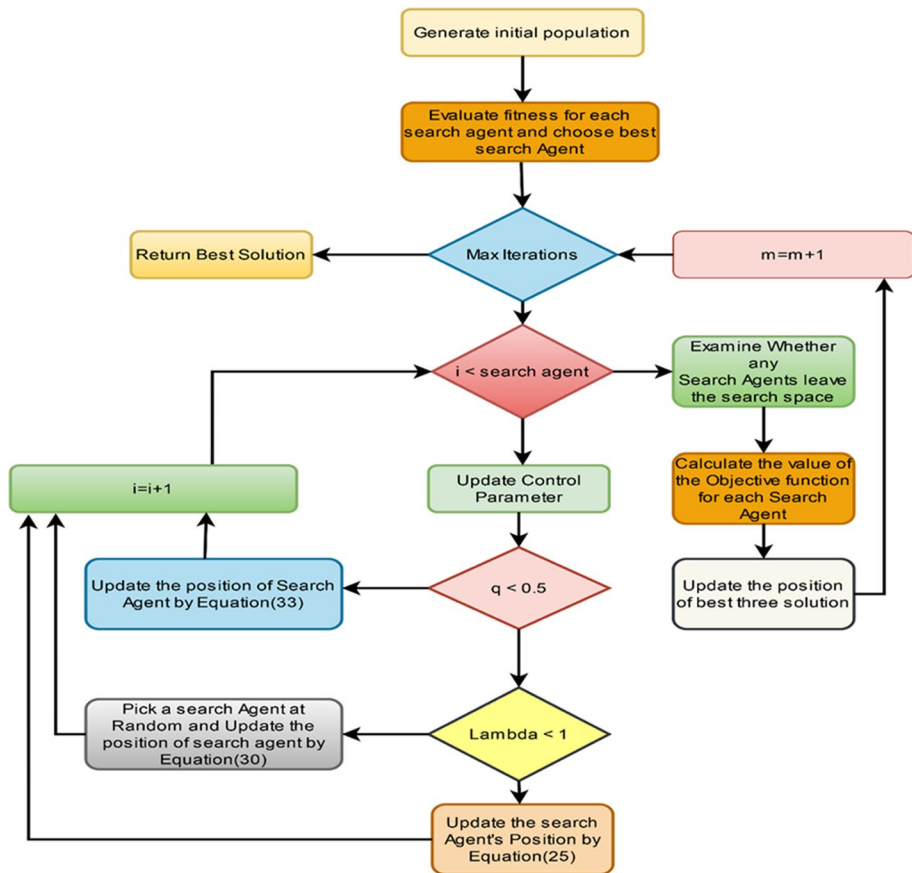


Fig. 5 Flowchart of the hGWWOA algorithm

$$O(\text{niter} \times \text{nwhales} \times D)$$

where niter is the number of iterations, nwhales is the number of whales, and D is the dimensionality of the problem. This means that the complexity of WOA increases linearly with these parameters. Therefore, if the problem size or the number of whales is large, the WOA algorithm may require more time and memory to find the optimal solution.

The time complexity of a hybrid algorithm that combines the GWO algorithm and the WOA would depend on the specific implementation details of the hybridization, including how the two algorithms are integrated and interact with each other. Generally, the time complexity of the hybrid algorithm would be influenced by the time complexities of GWO and WOA, as well as any additional computations required for the hybridization process. The hybridization involves running GWO and WOA sequentially or in some interlinked manner.

Time Complexity of GWO (per iteration): The time complexity of GWO's components, such as fitness evaluation, leader selection, and position update, would contribute to the overall complexity.

Time Complexity of WOA (per iteration): Similar to GWO, WOA's fitness evaluation, leader updates, and exploration strategies would contribute to its time complexity.

Hybridization overhead The process of combining GWO and WOA might involve extra calculations, data transfers, or synchronization steps. The complexity of these hybridization steps would depend on the specific approach taken.

The hybridization involves running both algorithms independently for a certain number of iterations each, the total time complexity would be the sum of their individual complexities:

$$\begin{aligned} \text{Total Time Complexity} \approx & (T_{\text{gwo}} * \text{GWO_Complexity}) \\ & + (T_{\text{woa}} * \text{WOA_Complexity}) \\ & + O(\text{hybridization_steps}) \end{aligned}$$

Where T_{gwo} is the number of GWO iterations, T_{woa} is the number of WOA iterations, GWO_Complexity is the time complexity of a GWO iteration, WOA_Complexity is the time complexity of a WOA iteration, and $O(\text{hybridization_steps})$ accounts for any additional computations due to the hybridization.

4 Results in tabular and graphical format

The dataset used in this study is ORIGA_ALL_FEATURES dataset which is our own customized dataset containing 65 extracted from 646 instances of ORIGA dataset. The Table 5, given below, provides the information about the dataset.

The dataset contains 646 instances i.e., 646 rows of data (Table 5). The dataset contains 66 columns, out of which the 65 columns are features and the last column is the classification column. The last column is the column with heading (glaucoma) which contains discrete data. The data is either 0 or 1. Here, 0 represents the person does not have glaucoma while 1 represents that the person has glaucoma. In 646 instances of data, we have 166 instances containing 1s while the other 480 instances contain 0s. All the experiments have been performed on a machine with Intel Core i5-8250U (8 Gen) at 1.60-1.80 GHz, 8 GB RAM and 64-bit Operating System. The language used for programming is Python (version 3.7) and the Jupyter Notebook platform is used for the same.

As previously indicated, we have successfully incorporated three bio-inspired algorithms into our system. Additionally, we have conducted a total of twelve tests, each utilizing a different algorithm. These tests involved manipulating the population size, specifically exploring variations of 5, 10, 15, and 20 populations. Furthermore, we conducted these tests over a span of 100 iterations, with subsequent intervals of 100 iterations, ultimately reaching a maximum of 500 iterations. In this study, we have successfully employed split-test 70:30 validation methodology to assess the efficacy of the proposed approach. When a subset of the features from the three bio-inspired algorithms are used in a test, there are eight ways to measure how well it works, such as statistical parameters, ROC curves, and execution time. These results are presented in Tables 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31 and 32. In addition to the aforementioned evaluations, we have also presented the

Table 5 Information about dataset

Dataset	No. of Instances	No. of Features
ORIGA_ALL_FEATURES	646	65

Table 6 Features selected and corresponding fitness value when experimented with population size 5 upto 500 iterations

No. of features	No. of iteration	Popula- tion size	Features Selected	Features number	Fitness value
65	100	5	34	1,2,5,6,8,11,13,14,16,21,22,23,24,25,25,26,28,30,31,32,35,36,37,39,40,43,44,47,50,51,52,54,55,61,64	0.009210
65	200	5	32	1,3,7,11,12,13,15,16,18,19,23,27,28,30,31,33,34,36,38,39,40,41,42,43,46,48,53,56,57,58,60,61	0.001331
65	300	5	33	0,1,2,5,8,12,13,16,17,21,23,25,26,28,30,32,33,34,35,37,41,42,44,46,49,52,54,55,58,60,61,62,64	0.000934
65	400	5	32	3,8,11,13,14,15,16,18,19,20,21,22,23,24,25,28,31,32,34,36,38,42,45,46,48,49,52,54,55,58,62,63	0.000922
65	500	5	30	1,4,5,6,9,10,11,21,24,28,29,30,31,32,33,35,38,39,41,43,45,46,47,48,49,59,60,61,63,64	0.000656

Table 7 Computed results for performance measuring metrics when experimented with population size 5 upto 500 iterations

	Accuracy	Precision	Sensitivity	Specificity	F1-score	Kappa score	MCC	AUC value	Execution time
Logistic Regression	0.912	0.939	0.951	0.809	0.943	0.763	0.761	0.887	0.473
XgBoost	0.930	0.950	0.960	0.848	0.968	0.810	0.829	0.908	0.722
Random Forest	0.955	0.961	0.990	0.889	0.982	0.888	0.891	0.957	0.790
CatBoost	0.955	0.969	0.982	0.871	0.976	0.875	0.875	0.945	9.129
SVM	0.934	0.936	0.977	0.800	0.954	0.807	0.801	0.921	0.589
Ensemble	0.943	0.957	0.976	0.859	0.963	0.846	0.853	0.931	9.268

Table 8 Features selected and corresponding fitness value when experimented with population size 10 upto 500 iterations

No. of features	No. of iteration	Population size	Features Selected	Features number	Fitness value
65	100	10	32	2,3,8,9,10,11,12,13,15,17,20,22,24,25,26,27,29,30,32,33,34,35,39,42,45,46,50,52,58,62,63,64	0.003249
65	200	10	30	0,9,12,13,14,15,16,20,22,24,26,29,30,33,35,36,41,42,45,46,48,49,51,54,55,57,58,60,62,63	0.000978
65	300	10	30	0,2,5,6,8,11,15,17,20,21,23,26,27,28,31,34,38,39,45,46,47,48,49,50,52,55,56,58,60,63	0.000901
65	400	10	28	1,2,3,4,7,10,13,14,15,18,19,20,24,28,30,40,41,44,46,47,48,50,51,53,54,57,59,62	0.000713
65	500	10	31	1,5,6,7,9,10,11,12,14,15,17,19,22,23,24,26,28,34,35,38,41,43,44,46,49,51,53,54,55,57,62	0.000299

Table 9 Computed results for performance measuring metrics when experimented with population size 10 upto 500 iterations

	Accuracy	Precision	Sensitivity	Specificity	F1-score	Kappa score	MCC	AUC value	Execution time
Logistic Regression	0.911	0.948	0.944	0.801	0.946	0.740	0.745	0.867	0.417
XgBoost	0.932	0.964	0.941	0.887	0.951	0.786	0.791	0.881	0.578
Random Forest	0.950	0.960	0.969	0.902	0.963	0.858	0.850	0.926	0.605
CatBoost	0.959	0.966	0.989	0.888	0.970	0.882	0.899	0.957	6.118
SVM	0.936	0.935	0.975	0.809	0.951	0.810	0.814	0.921	0.413
Ensemble	0.951	0.960	0.984	0.871	0.977	0.871	0.871	0.945	6.266

Table 10 Features selected and corresponding fitness value when experimented with population size 15 upto 500 iterations

No. of features	No. of iteration	Population size	Features Selected	Features number	Fitness value
65	100	15	29	0, 3, 5, 7, 8, 9, 15, 17, 22, 23, 24, 27, 28, 30, 32, 34, 35, 36, 38, 39, 42, 44, 45, 48, 50, 56, 58, 61, 64	0.003487
65	200	15	37	0, 2, 3, 5, 6, 9, 11, 13, 14, 15, 16, 17, 19, 20, 21, 22, 23, 25, 26, 31, 33, 40, 42, 43, 44, 46, 48, 49, 51, 52, 54, 55, 56, 58, 59, 60, 61	0.000816
65	300	15	32	4, 5, 9, 10, 11, 15, 21, 22, 23, 26, 27, 28, 32, 33, 34, 35, 36, 37, 38, 40, 41, 43, 44, 45, 46, 47, 53, 55, 57, 59, 60, 63	0.000534
65	400	15	31	1, 2, 3, 4, 5, 10, 11, 13, 18, 20, 21, 23, 24, 26, 32, 33, 34, 36, 38, 44, 45, 46, 48, 50, 52, 55, 56, 58, 59, 62, 63	0.000450
65	500	15	33	4, 5, 7, 9, 10, 13, 14, 17, 18, 20, 21, 25, 26, 27, 28, 29, 31, 35, 37, 38, 40, 44, 45, 48, 50, 51, 56, 57, 58, 61, 62, 63, 64	0.000235

Table 11 Computed results for performance measuring metrics when experimented with population size 15 upto 500 iterations

	Accuracy	Precision	Sensitivity	Specificity	F1-score	Kappa score	MCC	AUC value	Execution time
Logistic Regression	0.921	0.933	0.968	0.808	0.952	0.789	0.799	0.898	0.420
XgBoost	0.940	0.962	0.962	0.864	0.963	0.822	0.828	0.911	0.628
Random Forest	0.956	0.961	0.984	0.879	0.978	0.871	0.878	0.945	0.606
CatBoost	0.942	0.951	0.974	0.853	0.965	0.844	0.843	0.931	4.319
SVM	0.933	0.939	0.984	0.807	0.959	0.826	0.824	0.932	0.410
Ensemble	0.941	0.955	0.995	0.840	0.971	0.865	0.867	0.950	4.339

Table 12 Features selected and corresponding fitness value when experimented with population size 20 upto 500 iterations

No. of features	No. of iteration	Population size	Features Selected	Features number	Fitness value
65	100	20	30	0,1,3,5,8,9,11,12,13,14,17,20,21,23,24,25,30,31,33,36,40,41,44,46,49,50,51,58,59,63	0.001474
65	200	20	30	0,3,4,6,8,14,15,17,19,22,23,25,26,27,32,33,39,41,42,44,45,47,51,52,54,56,57,58,60,62	0.000645
65	300	20	31	0,1,3,4,7,8,10,11,15,16,21,23,24,26,31,32,33,37,39,41,42,43,44,45,48,49,50,57,62,63,64	0.000529
65	400	20	29	2,4,5,8,13,14,15,16,19,21,23,26,27,29,31,32,33,34,35,41,45,46,49,54,56,57,60,63,64	0.000285
65	500	20	33	1,3,6,7,8,9,14,17,18,19,22,26,27,29,31,35,37,39,41,43,44,45,47,49,50,51,56,57,59,60,61,62,64	0.000232

Table 13 Computed results for performance measuring metrics when experimented with population size 20 upto 500 iterations

	Accuracy	Precision	Sensitivity	Specificity	F1-score	Kappa score	MCC	AUC value	Execution time
Logistic Regression	0.931	0.941	0.979	0.812	0.964	0.824	0.835	0.924	0.571
XgBoost	0.942	0.953	0.971	0.859	0.962	0.843	0.842	0.931	0.997
Random Forest	0.955	0.954	0.982	0.852	0.963	0.862	0.872	0.9425	0.777
CatBoost	0.956	0.962	0.991	0.881	0.980	0.881	0.898	0.957	7.697
SVM	0.947	0.949	0.982	0.824	0.968	0.832	0.844	0.935	0.589
Ensemble	0.959	0.947	0.992	0.847	0.979	0.879	0.879	0.961	8.136

Table 14 Features selected and corresponding fitness value when experimented with population size 5 upto 500 iterations

No. of features	No. of iteration	Popula- tion size	Features Selected	Features number	Fitness value
65	100	5	33	3,6,7,9,10,16,18,19,20,22,24,25,29,31,34,35,37,40,41,45,47,49,50,51,52,53,54,55,56,57,59,61	0.08522
65	200	5	27	0,1,2,6,7,11,16,18,20,21,24,30,31,33,34,35,37,40,47,48, ,51,52,54,55,57,61,62	0.08396
65	300	5	39	1,2,3,4,7,8,9,10,12,13,14,15,16,18,19,20,22,23,24,29,31,33,36,37,40,41,42,43,44,45,49,50,53,54,57,59, 60,61	0.03665
65	400	5	37	1,2,3,7,8,9,11,12,16,17,18,19,20,21,22,23,25,26,28,29,30,34,37,38,39,40,43,46,49,50,51,52,53,56,58,5	0.02497
65	500	5	40	9,61 2,3,5,6,8,9,11,12,13,14,15,17,19,20,22,29,31,32,33,34,35,38,39,40,41,43,44,46,47,48,50,51,55,56,57,5	0.02356

Table 15 Computed results for performance measuring metrics when experimented with population size 5upto 500 iterations

	Accuracy	Precision	Sensitivity	Specificity	F1-score	Kappa score	MCC	AUC value	Execution time
Logistic Regression	0.917	0.933	0.958	0.791	0.945	0.772	0.773	0.898	0.663
XgBoost	0.932	0.967	0.953	0.860	0.956	0.807	0.807	0.900	0.838
Random Forest	0.953	0.953	0.986	0.857	0.969	0.872	0.875	0.953	1.044
CatBoost	0.948	0.953	0.979	0.854	0.966	0.857	0.858	0.942	6.585
SVM	0.932	0.926	0.985	0.792	0.955	0.821	0.828	0.940	0.497
Ensemble	0.938	0.941	0.979	0.829	0.959	0.831	0.834	0.935	6.566

Table 16 Features selected and corresponding fitness value when experimented with population size 10 upto 500 iterations

No. of features	No. of iteration	Population size	Features Selected	Features number	Fitness value
65	100	10	33	1,2,3,7,9,10,12,13,1,5,19,20,21,22,23,32,33,34,35,38,39,41,42,45,46,49,50,52,53,54,56,57,59,61	0.10489
65	200	10	26	6,7,8,10,12,16,17,20,23,37,38,41,42,44,45,46,47,50,53,55,56,57,59,60,61,63	0.06157
65	300	10	38	0,3,4,5,7,8,9,10,12,14,15,17,18,19,21,22,25,27,28,29,31,32,34,40,41,42,44,46,48,49,50,55,56,57,60,61,63	0.02342
65	400	10	31	1,2,5,6,9,10,11,12,13,17,19,20,21,23,24,25,28,30,33,35,36,42,43,45,46,53,54,56,58,60,63	0.01773
65	500	10	37	0,7,10,11,12,13,15,16,17,18,20,21,23,24,25,28,29,30,31,33,35,36,39,40,42,43,45,47,49,50,51,53,54,55,56,58,63	0.01261

Table 17 Computed results for performance measuring metrics when experimented with population size 10 upto 500 iterations

	Accuracy	Precision	Sensitivity	Specificity	F1-score	Kappa score	MCC	AUC value	Execution time
Logistic Regression	0.932	0.949	0.972	0.816	0.955	0.816	0.818	0.924	0.515
XgBoost	0.932	0.969	0.953	0.860	0.956	0.807	0.807	0.900	0.747
Random Forest	0.899	0.953	0.916	0.815	0.934	0.691	0.694	0.888	0.849
CatBoost	0.932	0.964	0.953	0.860	0.956	0.807	0.807	0.900	8.155
SVM	0.927	0.926	0.978	0.788	0.952	0.806	0.811	0.929	0.437
Ensemble	0.938	0.953	0.966	0.847	0.959	0.826	0.826	0.919	7.560

Table 18 Features selected and corresponding fitness value when experimented with population size 15 upto 500 iterations

No. of features	No. of iteration	Population size	Features Selected	Features number	Fitness value
65	100	15	33	0,1,5,6,11,12,14,15,16,18,19,20,21,23,27,4,35,37,38,39,40,42,43,44,45,46,47,48,52,53,57,63	0.04623
65	200	15	35	0,1,3,5,7,9,10,12,13,17,19,21,22,24,26,27,29,30,31,33,34,37,38,40,44,46,47,51,52,53,55,56,59,61,62	0.03860
65	300	15	27	0,7,8,10,12,13,21,22,25,26,30,32,33,35,36,37,40,41,42,43,46,48,51,56,58,59,61	0.02252
65	400	15	29	1,4,5,6,7,9,11,15,16,18,19,21,22,23,28,32,33,44,46,49,50,51,52,54,56,58,59,61,62	0.01874
65	500	15	37	1,2,6,10,11,15,18,21,22,24,26,27,28,31,32,33,34,36,39,41,43,47,50,56,58,61,62,63	0.0112

Table 19 Computed results for performance measuring metrics when experimented with population size 15 upto 500 iterations

	Accuracy	Precision	Sensitivity	Specificity	F1-score	Kappa score	MCC	AUC value	Execution time
Logistic Regression	0.927	0.933	0.972	0.809	0.953	0.803	0.806	0.921	0.609
XgBoost	0.948	0.953	0.979	0.854	0.966	0.857	0.858	0.942	0.820
Random Forest	0.932	0.953	0.959	0.844	0.956	0.810	0.810	0.908	0.784
CatBoost	0.932	0.953	0.959	0.844	0.956	0.810	0.810	0.908	6.091
SVM	0.927	0.926	0.978	0.788	0.952	0.806	0.811	0.929	0.482
Ensemble	0.953	0.953	0.986	0.857	0.969	0.872	0.875	0.953	7.171

Table 20 Features selected and corresponding fitness value when experimented with population size 20 upto 500 iterations

No. of features	No. of iteration	Population size	Features Selected	Features number	Fitness value
65	100	20	35	2,5,7,10,11,12,15,16,19,20,21,22,23,24,27,28,30,31,33,34,35, 42,43,44,49,50,51,52,56,57,58,59 ,60,61,62	0.07624
65	200	20	32	0,4,5,6,7,10,12,15,16,17,19,21,22,24,25,26,27,28,29,31,34,41,42,50,51,53,54,55,57,59,61	0.0249
65	300	20	33	4,5,6,8,11,12,14,16,19,20,21,23,25,26,27,31,32,33,34,36,41,42,44,48,49,50,51,52,53,59,60,61,6 2,	0.01815
65	400	20	35	0,1,4,5,10,12,14,15,17,18,19,21,22,23,26,27,29,31,35,36,39,41,42,43,46,48,49,50,51,53,56,59,6 1,62	0.01571
65	500	20	40	0,4,7,9,10,11,13,15,16,18,21,24,25,26,27,30,32,33,34,35,37,38,39,40,42,44,45,46,49,50,52,54,5 5,56,57,58,60,61,62,63	0.01241

Table 21 Computed results for performance measuring metrics when experimented with population size 20 upto 500 iterations

	Accuracy	Precision	Sensitivity	Specificity	F1-score	Kappa score	MCC	AUC value	Execution time
Logistic Regression	0.927	0.933	0.972	0.800	0.952	0.803	0.806	0.921	0.793
XgBoost	0.932	0.946	0.965	0.829	0.956	0.813	0.814	0.916	1.866
Random Forest	0.943	0.953	0.972	0.851	0.962	0.842	0.842	0.931	0.793
CatBoost	0.944	0.953	0.972	0.851	0.962	0.842	0.842	0.931	7.440
SVM	0.942	0.933	0.992	0.811	0.962	0.849	0.855	0.955	0.577
Ensemble	0.948	0.946	0.986	0.848	0.965	0.859	0.862	0.950	7.057

Table 22 Features selected and corresponding fitness value when experimented with population size 5 upto 500 iterations

No. of Features	No. of Iteration	Popula- tion Size	Features Selected	Features Number	Fitness Value
65	100	5	28	3, 6, 7, 11, 12, 15, 16, 18, 20, 21, 22, 23, 26, 28, 30, 33, 34, 38, 39, 40, 42, 45, 47, 49, 51, 53, 59, 64	1.3813e-1
65	200	5	25	3, 4, 5, 7, 11, 19, 21, 24, 25, 28, 29, 31, 32, 34, 39, 42, 43, 47, 49, 51, 55, 56, 60, 61, 63	6.4685e-2
65	300	5	33	0, 4, 5, 6, 10, 11, 12, 13, 15, 21, 22, 23, 24, 25, 28, 30, 31, 32, 33, 37, 38, 39, 44, 48, 50, 52, 53, 58, 59, 60, 61, 63, 64	8.7725e-3
65	400	5	34	0, 3, 4, 6, 7, 8, 9, 10, 12, 13, 15, 16, 17, 22, 23, 24, 25, 28, 29, 34, 36, 37, 40, 45, 46, 51, 53, 54, 57, 59, 61, 62, 63, 64	2.9202e-3
65	500	5	32	6, 8, 12, 15, 18, 19, 22, 26, 27, 28, 30, 31, 34, 35, 37, 38, 41, 42, 44, 45, 46, 47, 48, 49, 50, 51, 52, 57, 58, 60, 62, 63	1.7042e-4

Table 23 Computed results for performance measuring metrics when experimented with population size 5 upto 500 iterations

	Accuracy	Precision	Sensitivity	Specificity	F1-score	Kappa score	MCC	AUC value	Execution time
Logistic Regression	0.933	0.942	0.979	0.822	0.969	0.822	0.822	0.924	0.980
XG Boost	0.955	0.951	0.980	0.853	0.978	0.862	0.869	0.942	1.077
Random Forest	0.931	0.953	0.978	0.833	0.969	0.816	0.818	0.916	1.613
CAT Boost	0.943	0.954	0.972	0.851	0.963	0.849	0.843	0.931	14.853
SVM	0.941	0.941	0.987	0.825	0.966	0.830	0.841	0.935	1.091
Ensemble	0.968	0.957	0.990	0.863	0.971	0.895	0.898	0.965	15.101

Table 24 Features selected and corresponding fitness value when experimented with population size 10 upto 500 iterations

No. of Features	No. of Iteration	Popula- tion Size	Features Selected	Features Number	Fitness Value
65	100	10	32	1, 5, 9, 10, 11, 12, 17, 19, 22, 24, 25, 26, 27, 30, 31, 34, 37, 40, 41, 43, 45, 46, 48, 50, 51, 53, 54, 56, 58, 59, 62, 64	2.383e-1
65	200	10	34	1, 5, 6, 8, 10, 12, 13, 17, 20, 21, 26, 27, 28, 29, 31, 32, 33, 35, 36, 41, 42, 43, 44, 45, 46, 49, 50, 51, 53, 55, 58, 59, 60, 64	1.558e-2
65	300	10	30	0, 3, 7, 9, 10, 15, 17, 20, 22, 25, 26, 29, 30, 33, 34, 35, 36, 40, 41, 42, 43, 44, 45, 46, 49, 52, 58, 59, 63, 64	4.225e-3
65	400	10	34	1, 2, 3, 4, 6, 8, 9, 15, 16, 19, 20, 22, 23, 26, 27, 29, 30, 35, 36, 38, 39, 40, 41, 42, 45, 48, 50, 51, 53, 55, 60, 62, 63, 64	1.839e-3
65	500	10	31	2, 5, 6, 8, 10, 11, 12, 14, 15, 17, 18, 23, 26, 29, 31, 33, 34, 36, 38, 40, 42, 45, 48, 49, 52, 54, 55, 59, 60, 61, 64	2.383e-4

Table 25 Computed results for performance measuring metrics when experimented with population size 10 upto 500 iterations

	Accuracy	Precision	Sensitivity	Specificity	F1-score	Kappa score	MCC	AUC value	Execution time
Logistic Regression	0.939	0.942	0.972	0.808	0.955	0.809	0.814	0.921	0.927
XG Boost	0.959	0.951	0.981	0.869	0.976	0.877	0.883	0.953	1.143
Random Forest	0.967	0.963	0.990	0.883	0.985	0.905	0.901	0.968	1.038
CAT Boost	0.941	0.968	0.979	0.872	0.963	0.843	0.849	0.923	11.089
SVM	0.933	0.939	0.990	0.799	0.955	0.817	0.812	0.929	0.691
Ensemble	0.956	0.956	0.989	0.858	0.976	0.868	0.879	0.942	13.339

Table 26 Features selected and corresponding fitness value when experimented with population size 20 upto 500 iterations

No. of Features	No. of Iteration	Popula- tion Size	Features Selected	Features Number	Fitness Value
65	100	15	29	0, 1, 3, 7, 13, 14, 20, 21, 25, 27, 29, 31, 32, 33, 34, 39, 40, 43, 45, 46, 47, 48, 49, 50, 55, 56, 58, 61, 64	5.499e-2
65	200	15	27	0, 1, 2, 4, 6, 13, 16, 17, 18, 19, 21, 22, 23, 24, 26, 27, 28, 33, 34, 35, 39, 41, 44, 47, 48, 56, 64	3.624e-2
65	300	15	29	1, 2, 3, 5, 7, 8, 13, 15, 19, 22, 23, 24, 26, 27, 28, 30, 31, 32, 33, 35, 36, 38, 44, 47, 48, 51, 61, 62, 63	9.186e-3
65	400	15	33	0, 2, 5, 8, 10, 13, 14, 17, 18, 21, 22, 23, 27, 28, 33, 37, 40, 41, 43, 44, 46, 47, 48, 50, 52, 53, 54, 55, 57, 58, 62, 63, 64	2.717e-4
65	500	15	28	2, 4, 6, 7, 12, 13, 15, 16, 18, 19, 22, 23, 26, 29, 30, 32, 34, 39, 40, 43, 47, 49, 53, 56, 59, 62, 63, 64	9.004e-5

Table 27 Computed results for performance measuring metrics when experimented with population size 15 upto 500 iterations

	Accuracy	Precision	Sensitivity	Specificity	F1-score	Kappa score	MCC	AUC value	Execution time
Logistic Regression	0.927	0.944	0.795	0.799	0.952	0.797	0.791	0.909	0.795
XgBoost	0.965	0.965	0.993	0.881	0.979	0.892	0.899	0.957	1.073
Random Forest	0.945	0.951	0.988	0.842	0.966	0.851	0.855	0.939	1.048
CatBoost	0.956	0.951	0.985	0.854	0.972	0.863	0.864	0.942	10.439
SVM	0.925	0.937	0.973	0.786	0.958	0.799	0.792	0.917	0.951
Ensemble	0.948	0.958	0.981	0.842	0.962	0.851	0.851	0.939	10.930

Table 28 Features selected and corresponding fitness value when experimented with population size 20 upto 500 iterations

No. of Features	No. of Iteration	Popula- tion Size	Features Selected	Features Number	Fitness Value
65	100	20	35	0, 1, 4, 6, 7, 8, 11, 13, 14, 17, 22, 24, 25, 26, 30, 33, 36, 37, 39, 40, 41, 42, 45, 46, 47, 48, 50, 51, 52, 54, 55, 58, 59, 61, 62	3.945e-2
65	200	20	27	1, 2, 3, 4, 7, 9, 15, 16, 17, 18, 24, 26, 37, 40, 44, 45, 47, 49, 51, 52, 54, 56, 57, 59, 60, 63, 64	8.117e-3
65	300	20	29	1, 4, 7, 9, 17, 19, 20, 22, 23, 24, 29, 31, 36, 37, 40, 41, 42, 43, 44, 45, 48, 49, 53, 54, 57, 60, 61, 62, 64	1.262e-3
65	400	20	23	0, 4, 5, 6, 7, 9, 10, 12, 13, 16, 17, 20, 21, 26, 28, 32, 38, 49, 53, 54, 61, 63, 64	3.952e-4
65	500	20	24	1, 8, 9, 11, 12, 15, 17, 18, 20, 24, 26, 27, 28, 30, 32, 40, 41, 46, 47, 52, 54, 58, 61, 64	1.039e-4

Table 29 Computed results for performance measuring metrics when experimented with population size 15 upto 500 iterations

	Accuracy	Precision	Sensitivity	Specificity	F1-score	Kappa score	MCC	AUC value	Execution time
Logistic Regression	0.929	0.942	0.967	0.794	0.956	0.772	0.777	0.898	0.752
XgBoost	0.941	0.953	0.974	0.853	0.965	0.833	0.836	0.919	1.052
Random Forest	0.912	0.952	0.948	0.834	0.953	0.756	0.757	0.863	1.038
CatBoost	0.966	0.954	0.982	0.852	0.979	0.867	0.869	0.942	10.025
SVM	0.941	0.948	0.983	0.820	0.960	0.831	0.842	0.935	0.699
Ensemble	0.944	0.957	0.980	0.841	0.962	0.855	0.850	0.939	9.408

Table 30 Highest performance of various metrics computed using GWO algorithm

GWO algorithm's overall best results							
Specificity	Sensitivity	Precision	F1-score	Kappa score	MCC	AUC value	Accuracy
0.901 (Random Forest)	0.991 (CatBoost)	0.969 (CatBoost)	0.982 (Random Forest)	0.888 (Random Forest)	0.899 (CatBoost)	0.961 (Ensemble)	0.959 (CatBoost)

convergence curves (specifically the relationship between the number of iterations and the corresponding fitness values) (Figs. 6, 8, 10, and 12), illustrating the relationship between fitness and iterations for the GWO algorithm only (due to space constraints). Due to spatial limitations, the convergence curves have not been included for all tests conducted. Furthermore, we have included comprehensive tables (for all the experiments performed under three soft-computing algorithms) displaying the feature counts and fitness values. The corresponding performance metrics numerical results for GWO are shown using Tables 7, 9, 11, and 13. For WOA, these computed results are shown with the help of Tables 15, 17, 19, and 21, and their hybrid variants are depicted through Tables 23, 25, 27, and 29. Figures 7, 9, 11, and 13, 14, 15, 16, 17, 18, 19, 20 and 21 depict the confusion matrix for the best-performing classifier (best accuracy produced is the selection criteria) and combined receiver operating characteristic (ROC) curves derived from the conducted experiments. However, should anyone express interest in reviewing said data, it can be readily shared upon request. Furthermore, as a result of the aforementioned rationale, we are presenting the optimal values derived from all the experiments in Tables 30, 31, 32, and 33. The trials conducted have demonstrated the efficacy of the proposed methodology in accurately detecting the presence of this infection. Based on the empirical findings, it has been observed that the utilization of this particular methodology significantly augments the precision of classification in the context of glaucoma diagnosis, both in isolation and in conjunction with other approaches. The proposed methodology has the potential to aid healthcare professionals in rendering informed assessments regarding the diagnosis of glaucoma infection through the analysis of retinal fundus images. This, in turn, can optimize time utilization and enhance the efficacy of detection procedures.

4.1 Results of Algorithm-1 (GWO)

Here are the results of GWO when we have provided the dataset **ORIGA_ALL_FEATURES** to the particular code. We have performed the experiment on Population Size (5, 10, 15, and 20) with Iterations (100, 200, 300, 400, and 500). The results for the GWO are shown when the experiment performed on population 5 for iterations (100, 200, 300, 400, and 500). The column features selected shows the number of features that have been selected from the total features in the dataset at each individual iteration and the fitness value calculated at that iteration.

Table 31 Highest performance of various metrics computed using WOA algorithm

WOA algorithm's overall best results						
Specificity	Sensitivity	Precision	F1-score	Kappa score	MCC	AUC value
0.860 (XgBoost)	0.992 (SVM)	0.969 (XgBoost)	0.969 (Random Forest) and (Ensemble)	0.872 (Random Forest) and (Ensemble)	0.875 (Random Forest) and (Ensemble)	0.955 (SVM)
						0.953 (Random Forest) and (Ensemble)

Table 32 Highest performance of various metrics computed using hybrid algorithm

Hybrid algorithm's overall best results						
Specificity	Sensitivity	Precision	F1-score	Kappa score	MCC	AUC value
0.883 (Random Forest)	0.990 (Random Forest),(SVM), and (Ensemble)	0.968 (CatBoost)	0.979 (XgBoost) and (CatBoost)	0.905 (Random Forest)	0.901 (Random Forest)	0.968 (Random Forest) (Ensemble)

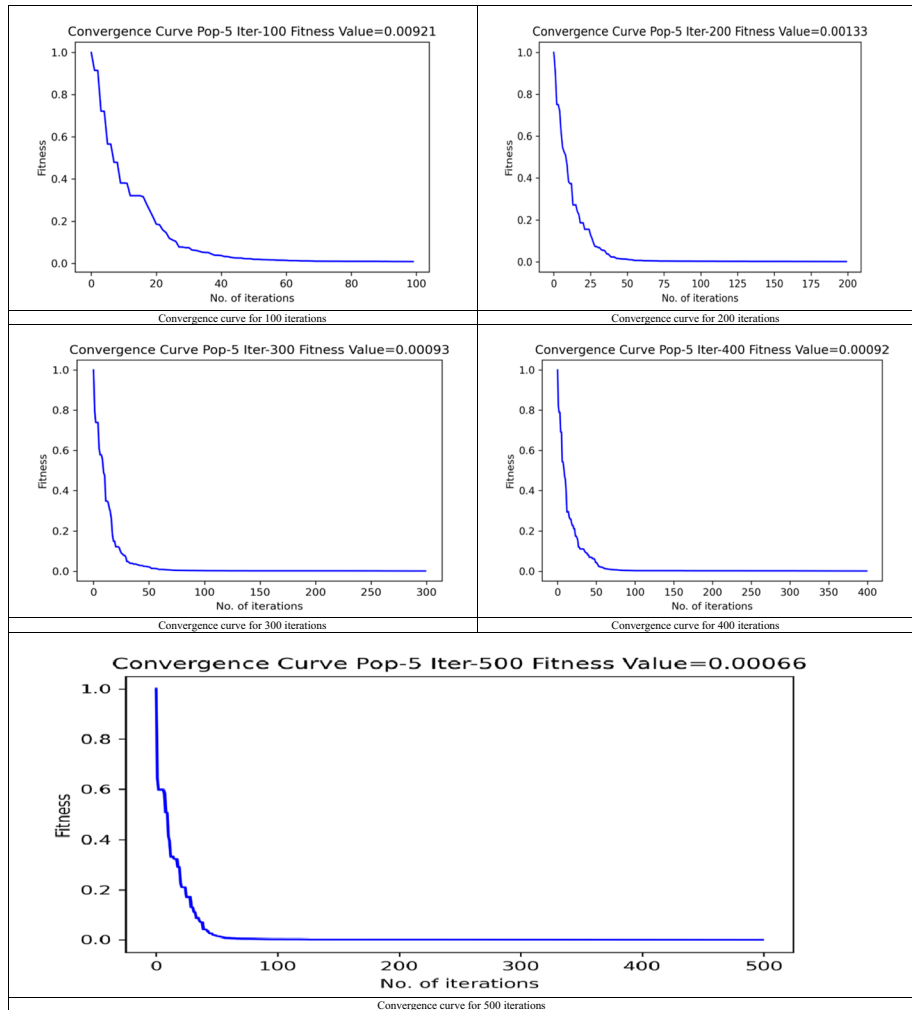


Fig. 6 Convergence curve for fitness vs iterations during population size 5

Convergence evaluation for population 5 The convergence behaviour of GWO algorithm is evaluated over the objective function (Rastrigin Function) with population 5 for Iterations (100, 200, 300, 400, 500) and the curves are depicted using Fig. 6.

Scores measured with Training-Testing-Splitting (70:30) with best confusion matrix and AUC curves are shown using Table 7 and Fig. 7 respectively.

The results for the GWO when the experiment performed on population 10 for iterations (100, 200, 300, 400, 500) are shown using Table 8. The column features selected shows the number of features that have been selected from the total features in the dataset at each individual iteration and the fitness value calculated at that iteration.

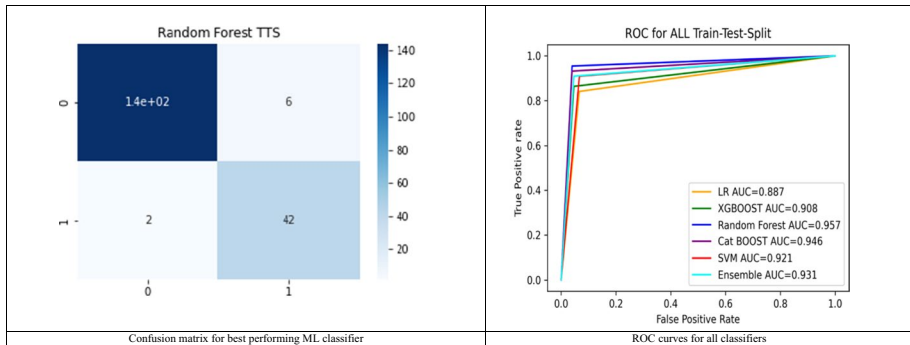


Fig. 7 Confusion matrix for best performing ML classifier and ROC curves during population size 5

Convergence evaluation for population 10 The convergence behaviour of GWO algorithm is evaluated over the objective function (Rastrigin Function) with Population 10 for Iterations (100, 200, 300, 400, and 500). The curves are depicted in Fig. 8.

Scores measured with Training-Testing-Splitting (70:30) with best confusion matrix and AUC Curve is shown with the help of Table 9 and Fig. 9.

Table 10 shows the results for the GWO when the experiment performed on population 15 for iterations (100, 200, 300, 400, and 500). The column features selected shows the number of features that have been selected from the total features in the dataset at each individual iteration and the fitness value calculated at that iteration.

Convergence evaluation for population 15 The convergence behaviour of GWO algorithm is evaluated over the objective function (Rastrigin Function) with population 15 for iterations (100, 200, 300, 400, and 500). The curves are shown using Fig. 10.

Scores measured with Training-Testing-Splitting (70:30) with best confusion matrix and AUC curves is presented using Table 11 and Fig. 11 respectively.

The results for the GWO when the experiment performed on population 20 for iterations (100, 200, 300, 400, 500) are revealed using Table 12. The column features selected shows the number of features that have been selected from the total features in the dataset at each individual iteration and the fitness value calculated at that iteration.

Convergence evaluation for population 20 The convergence behaviour of GWO algorithm is evaluated over the objective function (Rastrigin Function) with population 20 for iterations (100, 200, 300, 400, and 500) and the derived curves are shown using Figure 12.

Scores measured with Training-Testing-Splitting (70:30) with best confusion matrix and AUC curves are exposed with the help of Table 13 and Fig. 13.

4.2 Results of algorithm-2 (WOA)

Here are the results of Whale Optimization Algorithm when we have provided the dataset **ORIGA_ALL_FEATURES** to the particular code. We have performed the experiment on population size (5, 10, 15, and 20) with iterations (100, 200, 300, 400, and 500).

Table 33 Highest performance of various metrics computed during all algorithms

Specificity	Sensitivity	Precision	F1-score	Kappa score	MCC	AUC value	Accuracy
0.901 (GWO with Ran- dom Forest)	0.992 (WOA with SVM)	0.969 (GWO with Cat- Boost) and (WOA with XgBoost)	0.982 (GWO with Ran- dom Forest)	0.905 (Hybrid with Ran- dom Forest)	0.901 (Hybrid with Ran- dom Forest)	0.968 (Hybrid with Ran- dom Forest)	0.968 (Hybrid with Ensem- ble)

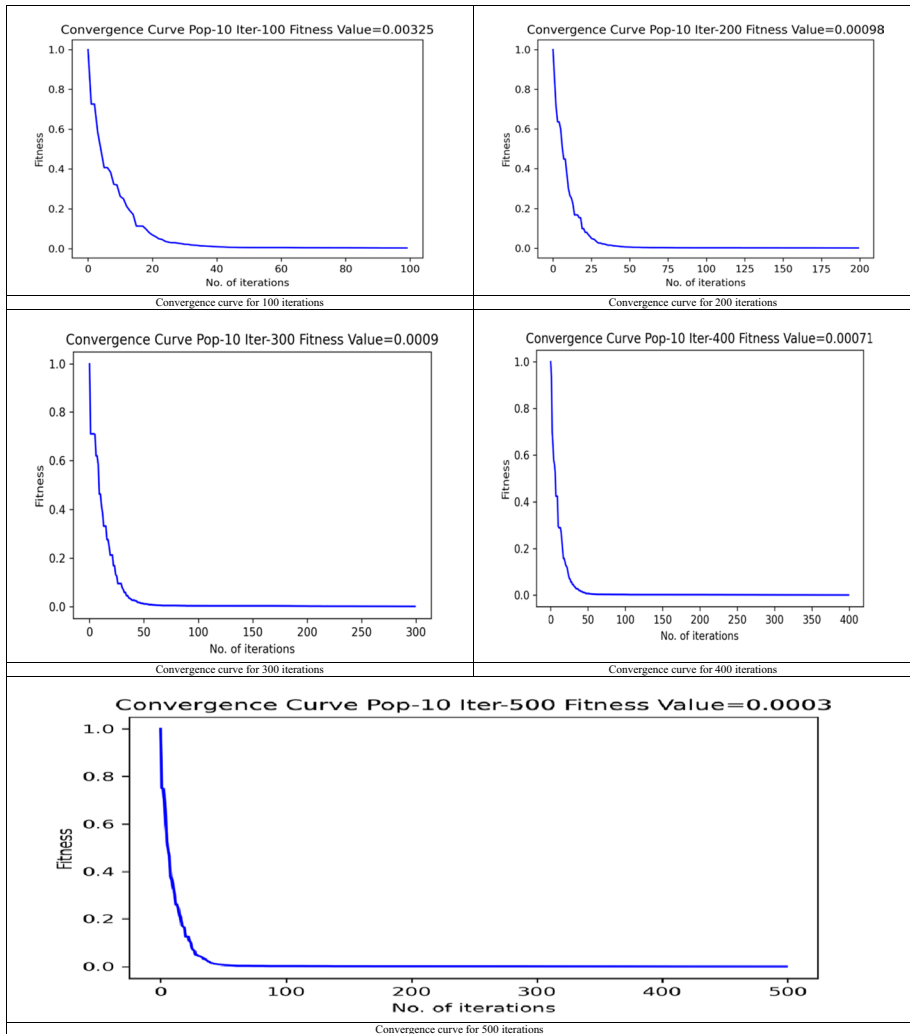


Fig. 8 Convergence curve for fitness vs iterations during population size 10

Due to space constraints convergence curves are not shown WOA and hGWWOA. If any reader is interested, the same can be shared personally. The results for the WOA when the experiment performed on population 5 for iterations (100, 200, 300, 400, 500) are revealed using Figure 14. The column features selected shows the number of features that have been selected from the total features in the dataset at each individual iteration and the fitness value calculated at that iteration.

Scores measured with Training-Testing-Splitting (70:30) with best confusion matrix and AUC Curves are exposed with the help of Table 15 and Fig. 14 respectively.

The results for the WOA when the experiment performed on population 10 for iterations (100, 200, 300, 400, 500) are revealed using Table 16. The column features selected shows the number of features that have been selected from the total features in the dataset at each individual iteration and the fitness value calculated at that iteration.

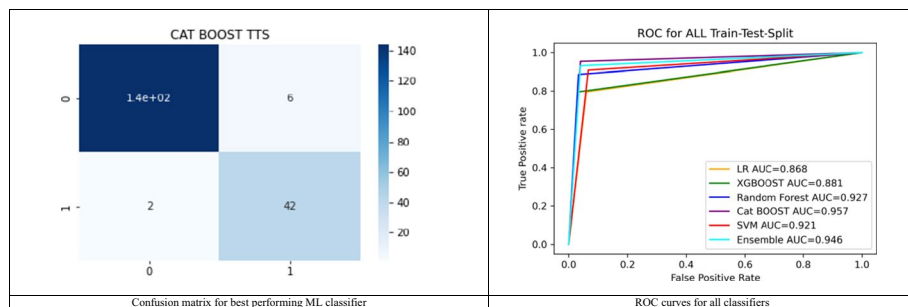


Fig. 9 Confusion matrix for best performing ML classifier and ROC curves during population size 10

Scores measured with Training-Testing-Splitting (70:30) with best confusion matrix and AUC Curves are exposed with the help of Table 17 and Fig. 15 respectively.

The results for the WOA when the experiment performed on population 15 for iterations (100, 200, 300, 400, 500) are revealed using Table 18. The column features selected shows the number of features that have been selected from the total features in the dataset at each individual iteration and the fitness value calculated at that iteration.

Scores measured with Training-Testing-Splitting (70:30) with best confusion matrix and AUC curves are exposed with the help of Table 19 and Fig. 16 respectively.

The results for the WOA when the experiment performed on population 20 for iterations (100, 200, 300, 400, 500) are revealed using Table 20. The column features selected shows the number of features that have been selected from the total features in the dataset at each individual iteration and the fitness value calculated at that iteration.

Scores measured with Training-Testing-Splitting (70:30) with best confusion matrix and AUC curves are exposed with the help of Table 21 and Fig. 17 respectively.

4.3 Results of algorithm-3 (Hybrid hGWWO)

Here are the results of hybrid algorithm when we have provided the dataset **ORIGA_ALL_FEATURES** to the particular code. We have performed the experiment on population size (5, 10, 15, 20) with Iterations (100, 200, 300, 400, 500). The results for the hybrid algorithm when the experiment performed on population 5 for iterations (100, 200, 300, 400, 500) are revealed using Table 22. The column features selected shows the number of features that have been selected from the total features in the dataset at each individual iteration and the fitness value calculated at that iteration.

Scores measured with Training-Testing-Splitting (70:30) with best confusion matrix and AUC curves are exposed with the help of Table 23 and Fig. 18 respectively.

The results for the hybrid algorithm when the experiment performed on population 10 for iterations (100, 200, 300, 400, 500) are revealed using Table 24. The column features selected shows the number of features that have been selected from the total features in the dataset at each individual iteration and the fitness value calculated at that iteration.

Scores measured with Training-Testing-Splitting (70:30) with best confusion matrix and AUC curves are exposed with the help of Table 25 and Fig. 19 respectively.

The results for the hybrid optimization algorithm when the experiment performed on population 15 for iterations (100, 200, 300, 400, 500) are revealed using Table 26. The column features selected shows the number of features that have been selected from the total features in the dataset at each individual iteration and the fitness value calculated at that iteration.

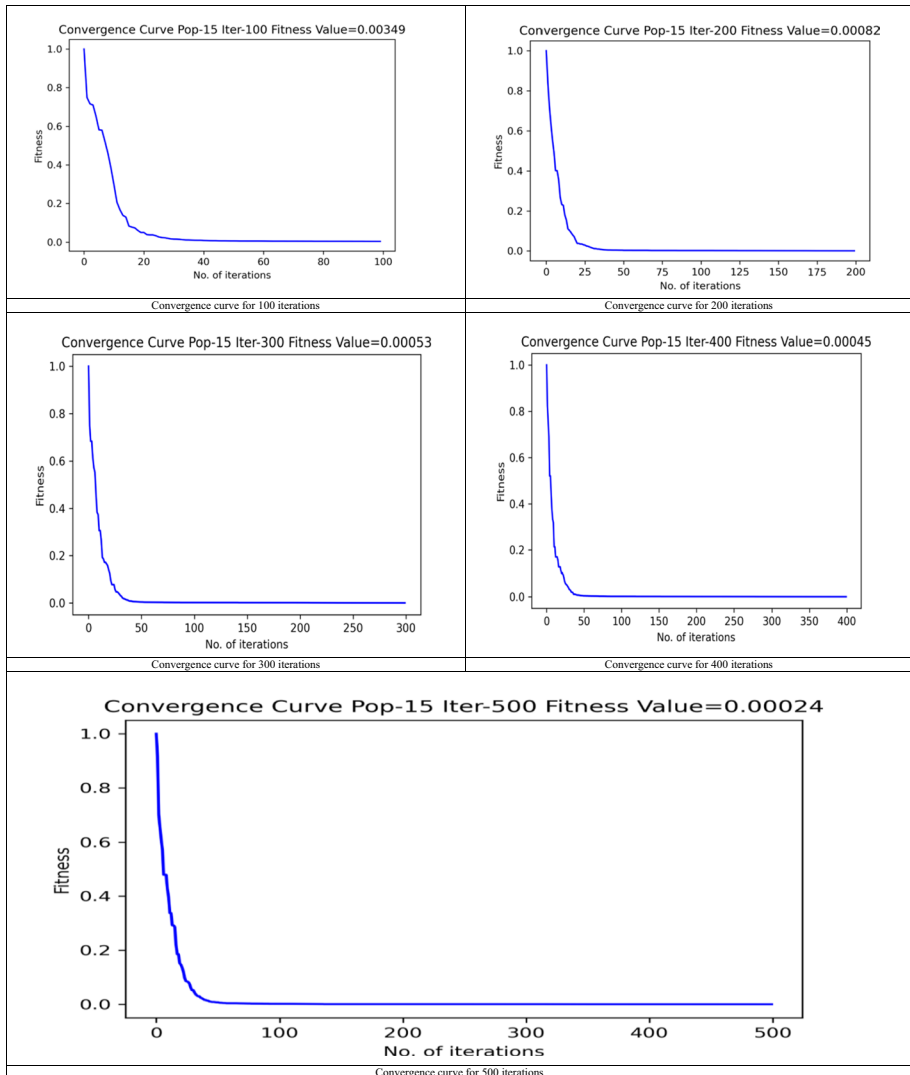


Fig. 10 Convergence curve for fitness vs iterations during population size 15

Scores measured with Training-Testing-Splitting (70-30) with best confusion matrix and AUC curves are exposed with the help of Table 27 and Fig. 20 respectively.

The results for the hybrid algorithm when the experiment performed on population size 20 for iterations (100, 200, 300, 400, 500) are revealed using Table 28. The column features selected shows the number of features that have been selected from the total features in the dataset at each individual iteration and the fitness value calculated at that iteration.

Scores measured with Training-Testing-Splitting (70:30) with best confusion matrix and AUC curves are exposed with the help of Table 27 and Fig. 20 respectively.

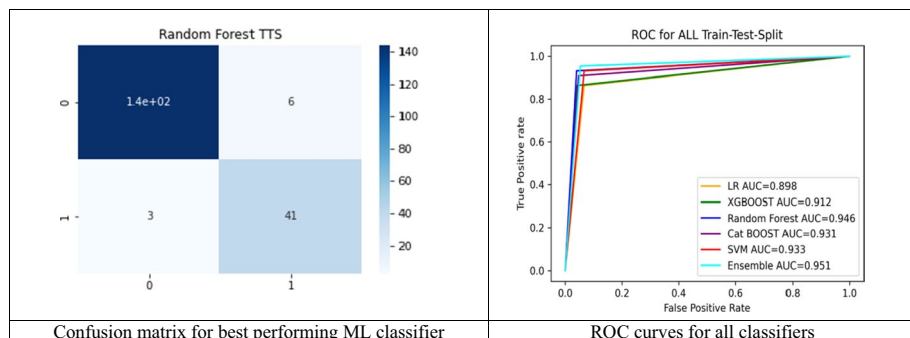


Fig. 11 Confusion matrix for best performing ML classifier and ROC curves during population size 15

Tables 30, 31, and 32 presents the highest performance achieved during all the experiments using GWO, WOA and their hybrid approach. Table 33 reveals the overall best results of the whole experiment process(all the combinations of soft-computing algorithms vs all implemented classifiers).

4.4 Discussion

The subsequent section will delve into a comprehensive analysis and examination of the findings and their far-reaching implications as derived from the study.

In recent years, there has been a notable escalation in the worldwide dissemination of this disease. Through the utilization of advanced image processing algorithms and cutting-edge artificial intelligence techniques, radiological images can be effectively harnessed to detect early indications of diseases. In recent academic research, scholars have conducted thorough investigations into the application of ML and DL models in the realm of early detection of glaucoma. The identification model that has been proposed leverages feature extraction and feature selection methods. It has exhibited promising classification performance when applied to fundus images, as evidenced in the aforementioned tables. Prior studies have also established that the application of a feature selection strategy based on soft computing techniques leads to a significant improvement in the performance of classification problems. The implications arising from these findings indicate that the application of feature selection methodologies holds promise in augmenting the efficacy of classification tasks. Furthermore, this methodology effectively mitigates costs related to time, a pivotal element in guaranteeing optimal patient care. The principal objective of this study is to develop and execute a proficient detection system that demonstrates enhanced performance and improved speed. This objective will be accomplished through the utilization of soft computing methodologies and the implementation of a machine learning-driven framework. Furthermore, the system will undergo evaluation utilizing a dataset of moderate proportions. The empirical investigations suggest that the proposed methodology demonstrated a notable degree of efficacy in the identification of glaucoma, as substantiated by the data presented in the aforementioned tables.

In recent years, significant advancements have been made in the utilization of artificial intelligence and ML methodologies for the purpose of disease prediction within human populations. Furthermore, the efficacy of ML methodologies is dependent upon the crucial

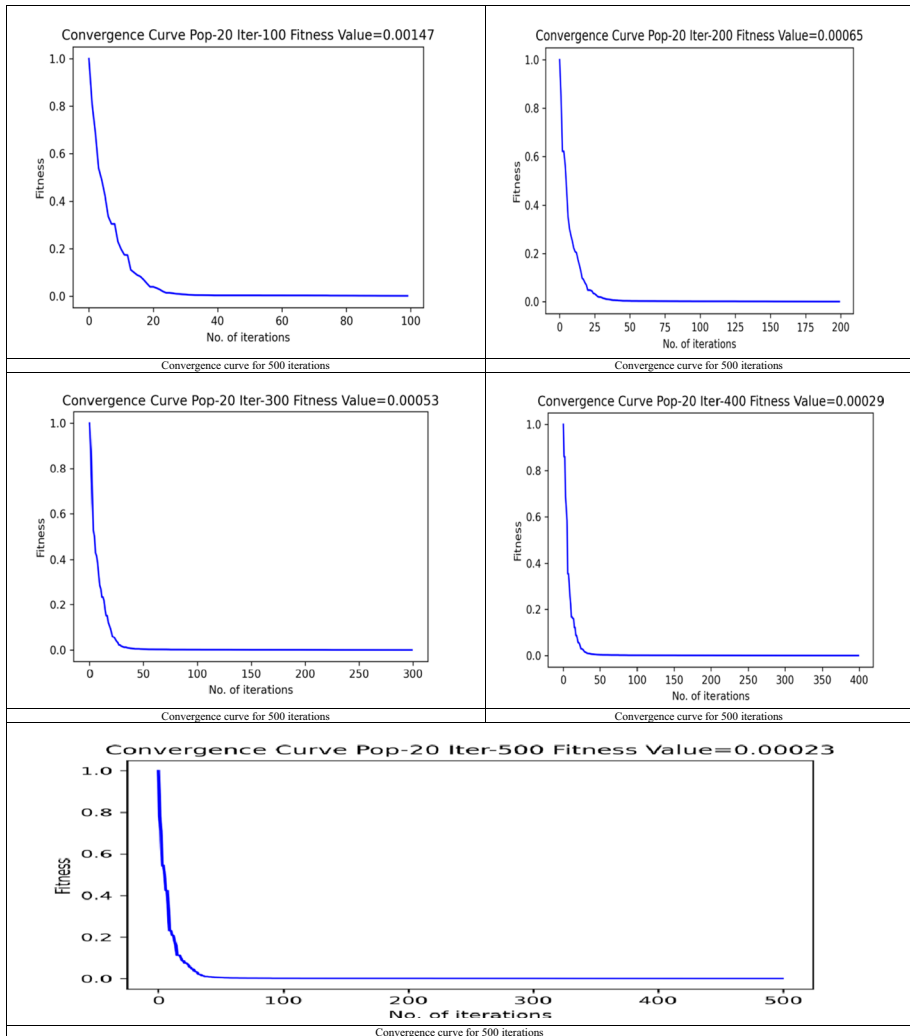


Fig. 12 Convergence curve for fitness vs iterations during population size 20

aspect of FS. The incorporation of FS techniques into ML algorithms for the prediction of glaucoma in patients holds significant promise in mitigating the temporal lag associated with medical testing results. This technological advancement would greatly enhance the ability of healthcare professionals to provide prompt and appropriate medical interventions to patients. This empirical study facilitates the advancement of an efficient CA-CDSS with the ability to accurately forecast the existence of glaucoma in patients, while minimizing dependence on human intervention and processing time. In order to advance the development of the proposed model, a careful selection of algorithms has been undertaken to effectively facilitate the training process of the glaucoma self-crafted dataset, specifically focusing on fundus images.

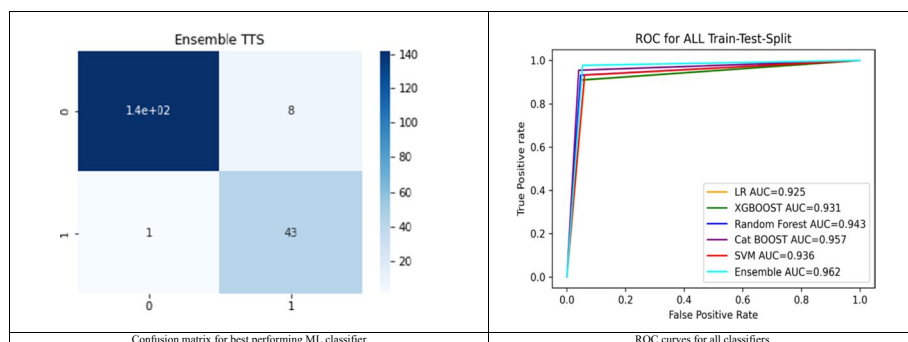


Fig. 13 Confusion matrix for best performing ML classifier and ROC curves during population size 20

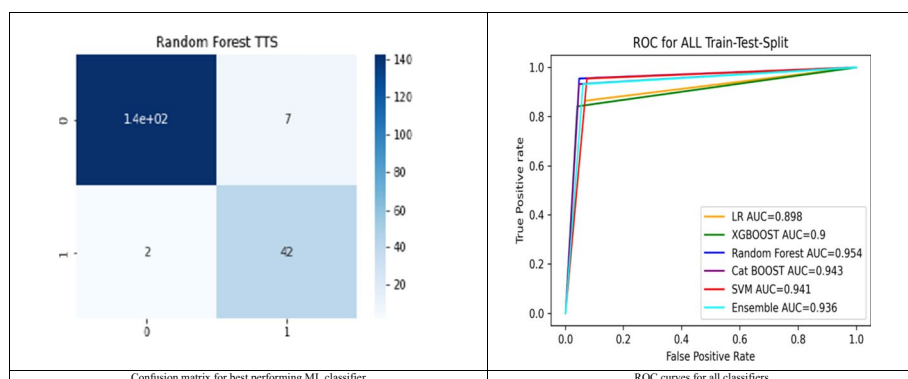


Fig. 14 Confusion matrix for best performing ML classifier and ROC curves during population size 5

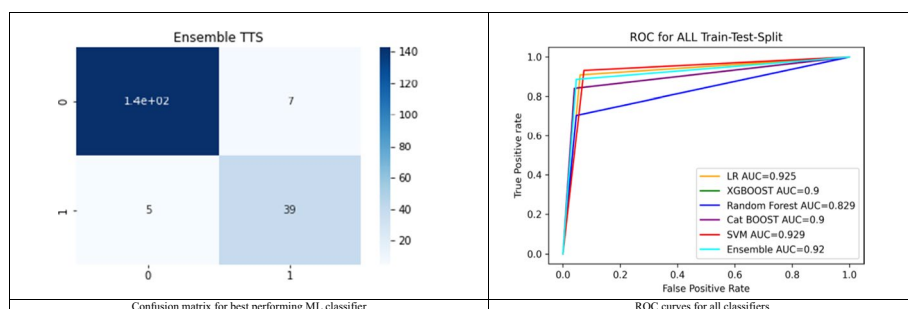


Fig. 15 Confusion matrix for best performing ML classifier and ROC curves during population size 10

The selected algorithms for this study encompass the GWO and WOA algorithm, both of which are recognized as highly efficient nature-inspired feature optimization algorithms. Following this, the dataset is employed for the purpose of training six ML models,

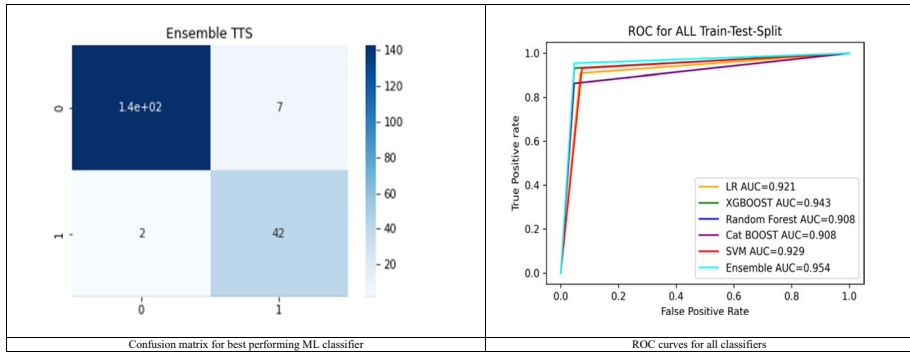


Fig. 16 Confusion matrix for best performing ML classifier and ROC curves during population size 15

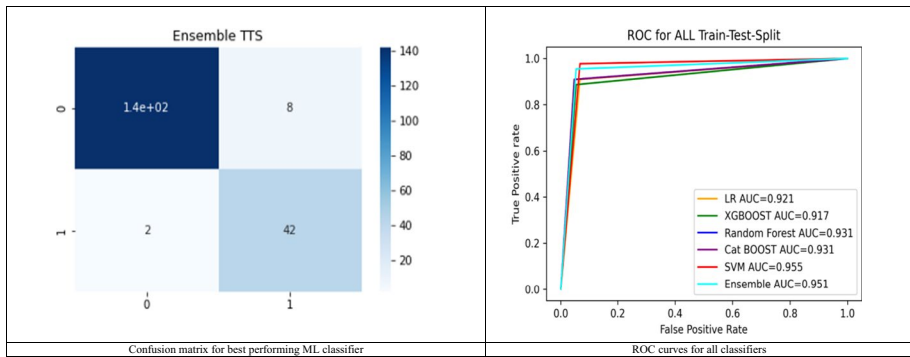


Fig. 17 Confusion matrix for best performing ML classifier and ROC curves during population size 20

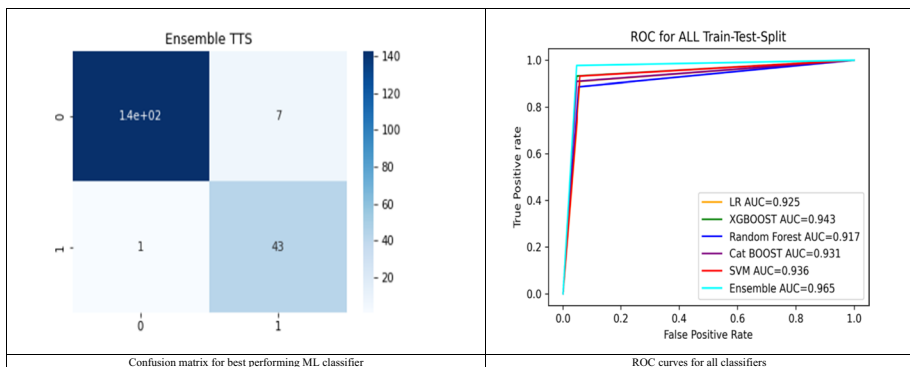


Fig. 18 Confusion matrix for best performing ML classifier and ROC curves during population size 5

specifically an ensemble model that amalgamates all of the five models. The proposed methodology encompasses three primary stages that are predicated upon the utilization of fundus images. During the preliminary phase, a code for self-programming is formulated

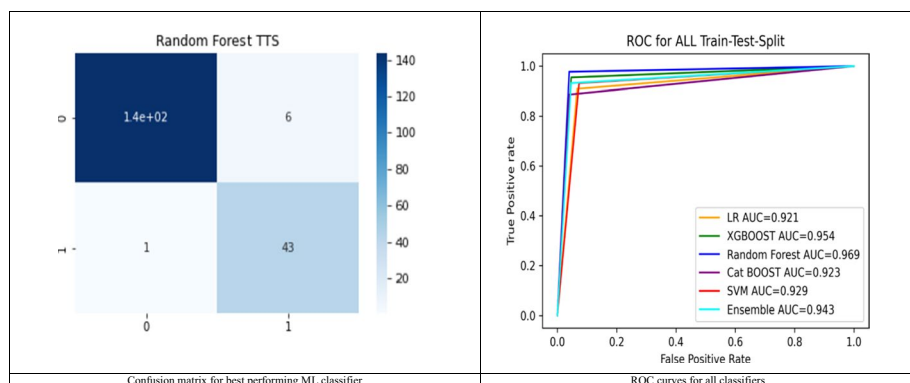


Fig. 19 Confusion matrix for best performing ML classifier and ROC curves during population size 10

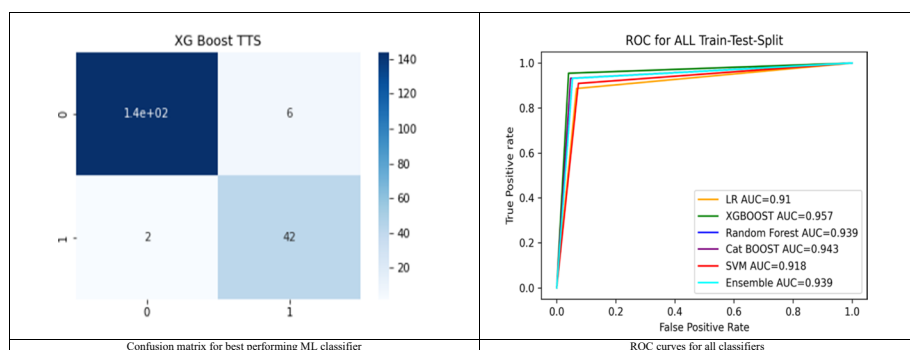


Fig. 20 Confusion matrix for best performing ML classifier and ROC curves during population size 15

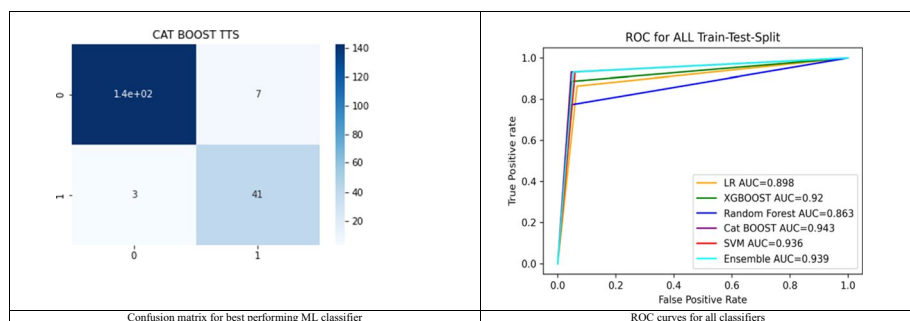


Fig. 21 Confusion matrix for best performing ML classifier and ROC curves during population size 20

with the objective of extracting features. During the subsequent phase, the selection process entails the identification of the most proficient characteristics from the initial set of extracted features. The categorization process entails the utilization of multiple ML classifiers, which

are employed to analyze pertinent features. This experimental study assesses and contrasts the performance of the selected algorithms with the intention of identifying the approach that demonstrates the utmost level of accuracy and optimal values for other efficiency metrics. The objective of generating feature significance values is to ascertain the degree to which these features exert influence on the prediction outcome. Fundus images have been recognized as a valuable diagnostic modality for the identification and evaluation of Glaucoma. The implementation of a computer-aided diagnostic methodology for the classification of Glaucoma, utilizing fundus images, holds significant promise in mitigating the workload of medical practitioners, in light of the widespread and still increasing infection cases. The conducted experiments entailed the utilization of split test (70:30) validation methodology. By utilizing the initial 65 features, a peak accuracy of 96.8% was attained. The successful execution of the proposed system delineated in this study holds the capacity to augment the decision-making processes of healthcare practitioners. The aforementioned system has exhibited notable advantages in terms of time efficiency, resulting in substantial time savings. Additionally, it has exhibited a commendable success rate in accurately predicting this illness through the utilization of fundus images.

After conducting an analysis of the all the trials, as presented in the aforementioned 33 tables, it has been observed that the Grey Wolf Optimization (GWO) algorithm demonstrates a maximum specificity value of 0.901 (when employing Random Forest). The CatBoost algorithm has determined that the maximum value of sensitivity for the other key factor is 0.991. The CatBoost algorithm yielded a maximum precision of 0.969; however the Random Forest algorithm yielded a maximum F1-score of 0.982. The accuracy achieved using CatBoost algorithm was 0.959, which can be deemed favorable. The receiver operating characteristic curve (ROC) analysis yielded an area under the curve (AUC) upto 0.959 (when employing CatBoost). Surprisingly, CatBoost has emerged as the most efficacious machine learning classifier, defying initial expectations of an ensemble methodology. Moreover, after conducting a thorough reassessment of the area under the curve (AUC) values, it is apparent that we consistently attained results surpassing 85% in every case, thereby demonstrating the remarkable effectiveness of the proposed methodology. The GWO optimization algorithm demonstrated remarkable outcomes for various parameters, including the Kappa score and MCC score, alongside the features it provided. The GWO algorithm effectively accomplished the task of reducing the initial set of 65 features to a minimum of 33 features of the original features, thereby yielding an approximate reduction of approximately 50% in the feature set. The algorithm's capacity to attain this outcome while upholding the performance of seven efficiency-measuring statistical parameters can be considered a noteworthy achievement.

After conducting a thorough analysis of the aforementioned 33 corresponding 24 tables, it can be deduced that the XGBoost algorithm demonstrates the highest level of specificity, reaching a value of 0.860 with the features selected by WOA. The sensitivity of the other major factor varies from a maximum value of 0.992 (as determined by SVM). The XGBoost algorithm yielded a maximum precision of 0.969. The Random Forest and Ensemble algorithms both achieved a maximum F1-score of 0.969. The Random Forest and Ensemble algorithms both yielded an accuracy of 0.953, indicating a commendable level of performance. The ROC demonstrated a diverse range of values, with the SVM classifier yielding the highest AUC value of 0.955, whereas the lowest observed value was 0.888. Furthermore, upon reevaluating the accuracy metric, it becomes apparent that achieving outcomes below 90% was consistently unattainable in all scenarios. This observation highlights the commendable performance of the recommended technique. The utilization of WOA optimization technique resulted in strong performance outcomes, as

evidenced by the robust results obtained from additional criteria such as Kappa and MCC scores. The WOA effectively achieved a reduction in the initial set by approximately 50% in the original feature set while maintaining the performance of seven efficiency-measuring statistical parameters.

The hybrid algorithm demonstrates a remarkable maximum specificity of 0.883 when leveraging the power of Random Forest. The determination is made through a comprehensive analysis of the data derived from the many trials as presented in the aforementioned 33 tables. The model exhibited a peak sensitivity of 0.9990 when employing the Random Forest, Support Vector Machine and Ensemble (in all three algorithms). The CatBoost model exhibited a peak precision value of 0.968. The XGBoost and CatBoost models produced an F1-score of 0.979 as the maximum value. Through the utilization of the Ensemble algorithm, our team has successfully attained a classification accuracy of 0.968, accompanied by a commendable minimum accuracy of 0.912. The ROC exhibited a range of values for the AUC, with the Random Forest algorithm achieving a remarkable score of 0.968. Moreover, upon reevaluating the precision, AUC values and other scores, it becomes apparent that we consistently attained an accuracy rate of no lower than 85%, signifying a noteworthy level of performance demonstrated by the proposed methodology. The hybrid optimization approach demonstrated outstanding performance in various additional metrics, such as the Kappa score and MCC score. The hybrid methodology effectively achieved a significant reduction in the initial set by 50% decrease. Remarkably, this reduction was accomplished without compromising the performance of the seven statistical metrics employed to assess efficiency. Hence, drawing upon the comprehensive examination of the aforementioned tables, it can be deduced that the proposed identification model, incorporating feature extraction and feature selection techniques, demonstrated commendable efficacy in the classification of fundus images.

5 Comparisons with prior published literature

Table 34 showcases a comprehensive comparative analysis between the proposed methodology and the presently prevalent diagnostic techniques for glaucoma. The table presented herein offers compelling evidence that the proposed innovative methodology demonstrates resilience and efficacy in precisely classifying fundus images, resulting in an auspicious accuracy rate of 96.8%. Similarly, the system exhibits favorable attributes in terms of both sensitivity and specificity in its performance. This serves as a demonstration of the overall applicability, expandability, and resilience of the proposed methodology. After conducting a comprehensive comparative analysis of our findings and comparing them with the outcomes of the remaining eleven methodologies published in or subsequent to 2018, as depicted in Table 34, it is evident that our performance stands out remarkably. However, it is imperative to acknowledge that there might exist discrepancies in the datasets employed for assessing alternative methodologies. Moreover, when compared to other recent publications in the field, the findings of this study exhibited a noteworthy improvement in the diagnostic effectiveness of retinal abnormalities linked to glaucoma. As a result, this medical assistance system possesses the capacity to effectively facilitate the development of accurate and cost-effective CA-CDSS, which can be utilized for comprehensive glaucoma screenings. Consequently, this system plays a crucial role in expediting the timely detection of this condition. The primary phase of the procedure mentioned in [22] entails employing convolutional neural networks (CNN) to discern and extract the optic disc from a retinal

Table 34 A comparative study of the proposed method's performance in comparison to other work for glaucoma diagnosis

Study	Methods/Features and/or classification methods	No. of images	ACC	SEN	SPE
[22] 2019	DL based RCNN	ORIGA	Un-disclosed	0.711	Un-disclosed
[23] 2018	RF+SMOTE	ORIGA	76.90	0.799	0.738
[24] 2019	DL based a semi-supervised semantic segmentation approach employing conditional GANs.	650 images of ORIGA dataset and 400 images from REFUGE dataset.	ORIGA 76.57 REFUGE 82.78	ORIGA 0.727 REFUGE 0.7	ORIGA 0.8041 REFUGE 0.956
[25] 2019	DL based DENet which is Disc-aware ensemble network,	ORIGA and Singapore Chinese Eye Study	84.29	0.847	0.838
[26] 2020	11 CNN models (P1) pretrained CNNs for transfer learning; (P2) pretrained deep CNNs as the feature extractors; (P3) an ensemble of methods, P1 and P2.	2787 images from ORIGA, ACRIMA, REFUGE and other datasets	From 80.00 (ORIGA) to 95.59 (REFUGE)	Un-disclosed	Un-disclosed
[27] 2020	UNet++ for segmentation from which clinical features were extracted. Increasing field of view (IFOV) was employed for extraction of image based features. Finally features were optimized. Gradient boosting decision tree (GBDT) was used for classification	650 images of ORIGA	84.3	0.894	0.793
[70] 2020	OD and OC Segmentation and then classification using DL based approach employing self crafted CNNs (MobileNetV2 feature extractor as a backbone.)	2482 images from ORIGA and other datasets	88	0.83	0.90
[71] 2021	Implemented Deep Learning based approach(CNN)	10658 images from REFUGE, and other datasets	95.3	0.841	0.958
[72] 2021	BEMD algorithm to decompose the ROI to components (BIMFs + residue). VGG-19 extracts features from decomposed BEMD components. Fuse the features of the same ROI and applying PCA to reduce features count.	ORIGA	96.67	0.966	0.967

Table 34 (continued)

Study	Methods/Features and/or classification methods	No. of images	ACC	SEN	SPE
[73] 2021	Implemented DL based approach by employing Self Crafted CNN	ORIGA	78.32	0.580	0.924
[74] 2021	Implemented DL based approach by employing Custom UNET++ model	650 images of ORIGA	90	0.87	0.81
Proposed method	65 features belonging to various categories. Feature selection using soft-computing approaches. Classification With six ML classifiers (Logistic Regression , XgBoost , SVM, Random Forest,CatBoost and Ensemble)	Public Dataset ORIGA	Highest 96.8	Highest 0.992	Highest 0.901

fundus image. In the subsequent phase, Deep CNNs are utilized to classify the extracted disc as either exhibiting a state of good health or being impacted by the ocular condition known as glaucoma [22]. The authors have presented a two-stage methodology for the localization and categorization of the optic disc (OD) in glaucoma screening. The initial implementation entailed the utilization of a fully automated disc localization methodology that relied on a faster region-based convolutional neural network (RCNN). Moreover, an investigation was conducted on the classification of images into diseased and healthy states by employing Convolutional Neural Networks (CNNs). In [24] practitioners have formulated a unique and discerning semi-supervised methodology for the purpose of semantic segmentation. This methodology utilizes conditional Generative Adversarial Networks (GANs) in order to accomplish the concurrent segmentation of the optic disc and cup [24]. The framework consists of three key components: a segmentation network, a generator, and a discriminator. In summary, the authors present a compelling rationale for the proposed methodology, asserting that it achieves state-of-the-art segmentation results on both the ORIGA and REFUGE datasets. Researchers have undertaken a comprehensive analysis of two innovative glaucoma detection methodologies, both of which leverage cutting-edge deep learning techniques [25]. The primary element encompasses a multi-label segmentation network, known as M-Net, which proficiently tackles the objective of concurrently segmenting the optic disc and optic cup. The process of glaucoma risk assessment entails the evaluation of the vertical CDR utilizing a segmented optic disc and cup. The second network, known as DENet, is an ensemble network that integrates the understanding of the optical disc area alongside the deep hierarchical context of the global fundus image. The DENet algorithm is capable of generating glaucoma detection results without the requirement of image segmentation. This study presents a coherent CAD framework for the identification of glaucoma using fundus images, which can be executed offline on mobile devices [70]. A meticulous amalgamation of numerous publicly accessible fundus photo datasets was undertaken with the objective of training CNNs for the tasks of segmentation and classification. Regarding the classification aspect, the obtained outcomes encompass an accuracy value of 0.87, a sensitivity value of 0.85, and an AUC value of 0.93. The authors provided a comprehensive description of the automated two-stage glaucoma screening approach [26]. The segmentation of the optic disc area was conducted using a DeepLabv3+ architecture. In this process, the encoder module was enhanced by replacing multiple deep CNNs. The researchers undertook an investigation into multiple proposed methodologies, employing a collection of five openly accessible datasets. This collection consisted of a grand total of 2787 retinal images. Based on their comprehensive analysis, it has been determined that the optimal approach for optic disc segmentation entails the synergistic utilization of DeepLabv3+ and MobileNet methodologies. The collection of methodologies exhibited exceptional performance in contrast to conventional techniques on the RIM-ONE, ORIGA, DRISHTI-GS1, and ACRIMA datasets, attaining accuracies of 97.37%, 90.00%, 86.84%, and 99.53% correspondingly. Furthermore, it is worth noting that the Area under Curve (AUC) values obtained for these datasets were as follows: 100%, 92.06%, 91.67%, and 99.98%, respectively. This study introduces an innovative methodology for the automated detection of glaucoma, integrating clinical assessment attributes with image-based features [27]. To attain accurate extraction of clinical measurement data, a proposition has been put forth to employ an enhanced UNet++ neural network for the simultaneous segmentation of the optic disc and optic cup. The process of segmentation is executed within a specified region of interest (ROI). The obtained segmentation results are employed to extract a range of essential clinical measurement data, which encompasses the calculation of the ratio between the optic cup and the optic disc. A gradient boosting

decision tree (GBDT) classifier has been developed with the primary aim of conducting glaucoma screening. The proposed methodology for ORIGA and DRISHTI-GS1 demonstrates a sensitivity value of 0.894, an accuracy value of 0.843, and an AUC value of 0.901. In the course of this research undertaking, a classifier was trained employing approximately 66.67% of the photographs [71]. A portion amounting to one-third of the photographs was designated for the explicit purpose of generating cross-validation assessment sets. The aforementioned sets have been meticulously crafted to replicate a screening scenario that closely emulates the prevalence of glaucoma in a highly realistic manner. The aim of this study was to assess the classifier's performance within the specific context of this scenario. The study revealed a prevalence rate of 24% for glaucoma. A total of fifty sets, each comprising one thousand images, were generated in a randomized fashion from the validation set. It is noteworthy that each set exhibited a glaucoma prevalence rate of 5%. The classifier exhibited an 84.1% detection rate and a 95.8% specificity rate when evaluated on the provided datasets.

6 Limitations and future scope

A significant limitation of this study is its exclusive reliance on a single dataset for the purpose of performance evaluation. However, it is possible to further expand the scope of this study by applying the proposed methodology to additional datasets. Additionally, exploring combinations of different datasets could also be considered for future research. The quantity of extracted features has the potential to be further augmented. Moreover, in the current study, we refrain from pre-processing the image to eliminate blood vessels. Our objective is to examine the impact of this procedure on the ultimate classification outcome. Another crucial aspect to consider is the classification step. Given that our main goal was to achieve efficient feature selection, the evaluated classifiers were not optimally fine-tuned. We are of the opinion that the enhancement of our results could be achieved through the utilization of multi-objective approaches for the purpose of fine parameter tuning.

Glaucoma is a multifaceted systemic condition. Hence, a comprehensive analysis of the visual system has the potential to enhance the prognosis and diagnosis of glaucoma. This objective can be accomplished by leveraging the advancements in neuro-imaging and retinal imaging technologies, which enable the integration of data from different components of the visual system. Potential future research in this field could involve exploring alternative feature selection techniques and employing diverse classification methodologies in order to enhance both the accuracy and efficiency of the results. In future research, it is recommended to explore the utilization of different features or the integration of texture, shape, and color features to address misclassifications and enhance the overall accuracy of the system. The evaluation of the performance of selected features using different statistical feature selection approaches remains an unresolved issue. Additionally, the development of a two-phase feature selection strategy, involving initial selection through classical statistical methods followed by further selection using soft-computing approaches, represents a novel research direction and presents an evaluative challenge. Another challenging issue involves the evaluation of performance for different variations of the Grey Wolf Optimizer (GWO) and Whale Optimization Algorithm (WOA), as suggested by scholars in recent times. Moreover, researchers are currently developing a cutting-edge soft-computing-based optimization algorithm with innovative features. Therefore, by leveraging a more robust integration of these recently introduced algorithms with highly efficient classifiers, there

is potential to enhance the accuracy of classification even further. The issue at hand can also be conceptualized and transformed into a multi-objective problem, wherein the two conflicting objectives are to maximize accuracy and minimize the number of selected features. The ongoing collaboration between researchers in the field of machine learning and ophthalmologists will remain of utmost importance. The integration of domain knowledge and technical expertise has the potential to yield more efficient and impactful solutions. Additionally, in the foreseeable future, there is potential to expand the scope of this work by incorporating automation techniques, specifically leveraging deep learning methodologies and a more extensive dataset, in order to enhance accuracy. The present approach can be further expanded in subsequent endeavors to seamlessly integrate with the graphic user interface designed for glaucoma diagnosis. Additionally, it can encompass the capability to identify and discern other ocular ailments, including retinal hemorrhage, macular edema, and diabetic retinopathy.

7 Conclusion

A novel methodology has been employed to identify the presence of glaucoma in fundus images. A total of sixty-five distinct features, each categorized into different classes, have been extracted from the subject images. Three soft computing-based algorithms have been utilized to achieve optimal feature selection in a highly efficient manner. The classification process has been executed utilizing a hybrid feature subset, employing various ML classifiers. The proposed methodology demonstrates a commendable accuracy rate of 96.8% in the diagnosis of glaucoma, employing an ensemble technique. This performance surpasses that of alternative classifiers, further validating the efficacy of the designed approach. This substantiates the high potential of our hybrid approach when leveraging selected features, even in the context of mid-size dataset. The empirical evidence suggests that the proposed methodology demonstrates a high level of effectiveness and robustness. Based on the outcomes of our comprehensive investigations, we have successfully compiled a concise compilation of distinctive features, thereby bestowing a significant advantage upon the esteemed researcher community. The proposed technique's exceptional accuracy and low feature requirements support its superiority to competing relevant methodologies. This research study provides a substantial contribution to the advancement of human society by serving as an initial support system for patients residing in countries facing a scarcity of medical practitioners. Additionally, it has the potential to serve as a valuable tool for ophthalmologists engaged in large-scale clinical testing for the purpose of glaucoma diagnosis, offering a valuable second opinion. The proposed computerized glaucoma prediction approach is anticipated to effectively mitigate the workload of ophthalmologists.

Declarations

Data will be made available on reasonable request.

This manuscript is the authors' original work and has not been published elsewhere.

All authors have checked the manuscript and have agreed to the submission.

We also declare that we do not have any conflict of interest.

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The dataset analysed during the current study are available in the internet repository, and can also be made available from the corresponding author on reasonable request

Conflict of interest The authors declare that they have no conflict of interest.

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Authors and Affiliations

Law Kumar Singh¹ · Munish Khanna² · Shankar Thawkar³ · Rekha Singh⁴

✉ Law Kumar Singh
lawkumarcs@gmail.com

Munish Khanna
munishkhanna.official@rocketmail.com

Shankar Thawkar
shankarthawkar@gmail.com

Rekha Singh
singh.rekha70@gmail.com

¹ Department of Computer Engineering and Applications, GLA University, Mathura 281406, India

² Department of Computer Science and Engineering, Hindustan College of Science and Technology, Mathura 281122, India

³ Department of Information Technology, Hindustan College of Science and Technology, Mathura 281122, India

⁴ Department of Physics, Uttar Pradesh Rajarshi Tandon Open University, Prayagraj, Uttar Pradesh 211021, India